



EXCALIBUR



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EXCALIBUR — EXplainable and demoCrAtized pipeLInes for unBiased, trUstworthy, and Responsible human-centric AI

D2.1

Report on Trustworthy AI Principles

Work Package 2 — Task 2.1

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Executive Summary

Deliverable D2.1 establishes the conceptual and operational foundation for Trustworthy AI in EXCALIBUR. In accordance with Task 2.1, it focuses on Transparency, Fairness, and Accountability and translates regulatory, standardisation, governance, and scientific evidence into technical and auditable elements that can support subsequent EXCALIBUR research, implementation, and validation activities.

The regulatory and standardisation review was updated through August 2026. The initial landscape comprises 21 instruments (Table 1), of which five were selected for deeper operational analysis (Table 2). The structured Task 2.1 scientific-literature corpus comprises 18 original publications and four supplementary scientific sources formally incorporated in Appendix B. Together, these two evidence streams support the mapping from high-level principles, provisions, requirements, and guidance to metrics, controls, evaluation procedures, evidence artefacts, and interpretation criteria.

The principal contribution of D2.1 is a traceable distinction between (i) instrument-defined provisions and controls, (ii) literature-supported technical operationalisations, and (iii) project-defined candidate interpretation rules for later EXCALIBUR experimentation. The resulting metrics-and-thresholds mappings are presented in Tables 6–8, while Table 9 consolidates the EXCALIBUR-oriented Trustworthy AI requirements. The deliverable does not constitute a legal-compliance framework and does not infer compliance from a metric value or from the presence of a documentation artefact alone.

D2.1 therefore provides the evidence baseline carried forward to WP3, WP4, and WP5. It specifies what should be measured, documented, traced, and validated, while leaving the design of LLM-based explainability and fairness methodologies to WP3, their platform-level implementation to WP4, and their technical and human-centred validation to WP5.

D2.1 — Report on Trustworthy AI Principles

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Abbreviations

AEDT	Automated Employment Decision Tool (NYC Local Law 144)
ADHF	Algorithmic Design History File (SMACTR artefact)
AI Act	EU Regulation (EU) 2024/1689 on Artificial Intelligence
AIMS	Artificial Intelligence Management System (ISO/IEC 42001)
DI	Disparate Impact (selection-rate ratio)
FMEA	Failure Modes and Effects Analysis
FPR	False Positive Rate
GDPR	General Data Protection Regulation
GPAI	General-Purpose AI (model)
ISO/IEC	International Organization for Standardization / International Electrotechnical Commission
LIME	Local Interpretable Model-agnostic Explanations
ML	Machine Learning
NIST	U.S. National Institute of Standards and Technology
NPV	Negative Predictive Value
OMB	U.S. Office of Management and Budget
PPV	Positive Predictive Value
PRD	Product Requirements Document (SMACTR artefact)
RMF	Risk Management Framework (NIST AI RMF)
SHAP	SHapley Additive exPlanations
SMACTR	Scoping, Mapping, Artifact Collection, Testing, and Reflection (internal audit framework)
XAI	Explainable AI

1 Introduction

1.1 Trustworthy AI: from principles to practice

AI systems now make or support decisions in domains where errors may directly affect individuals, organisations, and society, such as clinical screening, hiring, credit scoring, judicial risk assessment, and health-related inference from wearable sensors [1, 2, 3]. As these systems have moved from research prototypes to deployed infrastructure, several recurring failure modes have become visible. AI models may operate opaquely, producing outputs through mechanisms that cannot be easily inspected or audited by the people their outputs affect [4, 5]. They may exhibit statistical bias that systematically disadvantages protected groups, often in ways that remain invisible without targeted measurement [3, 6, 7, 20, 24]. When outcomes are challenged, responsibility may also be diffused across data providers, model developers, deployers, and other actors, without standardised evidence that allows a third party to reconstruct what occurred and why. Generative AI systems introduce additional concerns, including hallucinations, toxic or biased outputs, and increased risks of misinformation, which remain only partially specified at the level of technical implementation guidance.

The response from policy bodies and governance initiatives [8, 9, 49, 50], standardisation organisations [38, 51], and the research community [21, 23, 25] has converged on a vocabulary of trustworthy or responsible AI organised around recurring principles, prominently including Transparency, Fairness, and Accountability. Major initiatives such as the AI-HLEG Ethics Guidelines [8], the OECD AI Principles [50], the NIST AI Risk Management Framework [49], and the EU AI Act [9] share related high-level goals but differ in scope, legal status, terminology, and level of operational specificity. A developer or evaluator reading these instruments in parallel may therefore encounter the same terms used with different operational meanings, similar goals pursued through different technical approaches [21, 23, 25], and provisions or recommendations that do not directly prescribe technical controls, evaluation criteria, or implementation requirements.

EXCALIBUR addresses this gap by translating regulatory and conceptual expectations into technical and evidence-based elements that can guide explainability and fairness assessment in AI pipelines. In line with Task 2.1, D2.1 acts as an initial translation layer between Trustworthy AI principles and the technical framework required by subsequent EXCALIBUR research and development activities. It maps relevant regulatory, standardisation, governance, and normative instruments against the three Task 2.1 pillars — Transparency, Fairness, and Accountability — and complements this mapping with scientific literature to identify suitable metrics, controls, evaluation procedures, documentation artefacts, thresholds, and interpretation criteria.

1.2 Focus of D2.1: Transparency, Fairness, and Accountability

The AI-HLEG Ethics Guidelines and related Trustworthy AI initiatives identify a broader set of requirements and principles, including human agency and oversight, technical robustness and safety, privacy and data governance, transparency, diversity/non-discrimination/fairness, societal and environmental well-being, and accountability [8]. In accordance with Task 2.1, D2.1 focuses on three dimensions that are directly connected to the EXCALIBUR framework: Transparency, Fairness, and Accountability.

In this deliverable, a distinction is made between regulatory transparency and technical explainability. Transparency is used as the broader regulatory and governance concept, encompassing disclosure, documentation, traceability, and the availability of information to relevant actors. Explainability/XAI denotes the technical methods and evaluation criteria used to make AI model behaviour and outputs understandable to relevant stakeholders. This distinction is formalised in §3.1 and applied consistently throughout the mapping.

Within the Transparency pillar, EXCALIBUR's principal technical interest lies in the explainability dimension — XAI methods and their evaluation as applied to model behaviour and individual outputs. Process-side transparency,

including disclosure, documentation, and record-keeping, remains necessary to the mapping because it provides the evidence and governance context within which technical explanations are produced and assessed. Accordingly, the term Transparency is retained for the broader pillar, while XAI is used when the analysis specifically concerns technical explainability methods or explanation-quality evaluation.

This focus does not imply that the remaining Trustworthy AI dimensions are irrelevant. Rather, they act as cross-cutting constraints on the implementation of the three main pillars. Human oversight is connected to the usability and contestability of explanations; privacy and data governance constrain the collection and processing of protected or sensitive attributes for fairness assessment; and safety and robustness affect the reliability of explanations and fairness evaluations. A more detailed treatment of Human Oversight, Societal Well-being, Safety, Privacy, Data Governance, and ethical compliance falls within Tasks 2.2 and 2.3 and is consolidated in Deliverable D2.2. D2.1 therefore remains focused on the Trustworthy AI principles mapping assigned to Task 2.1.

Operational definitions of the three pillars are stated in §3 and applied consistently throughout the analysis. Briefly:

- Transparency covers both the regulatory / process side — including disclosure, documentation, record-keeping, and traceability — and the technical explainability side — including interpretability and explanation of model behaviour and outputs.
- Fairness covers group, individual, and subgroup fairness, with worldview commitments — such as “We’re All Equal” (WAE) and “What You See Is What You Get” (WYSIWYG) — made explicit because they inform the appropriateness and interpretation of different metric families [23].
- Accountability covers auditability, responsibility allocation, governance processes, and the evidence artefacts that allow relevant actors to assess whether transparency and fairness requirements have been considered, implemented, and documented.

Where the technical and legal vocabularies diverge, §3 records the divergence and adopts operational meanings that are suitable for the mapping objectives of D2.1.

1.3 Problems Addressed

The rapid development and deployment of AI systems has created a persistent gap between high-level Trustworthy AI principles and their practical technical assessment. Regulatory, standardisation, governance, and normative instruments increasingly establish expectations concerning transparency, fairness, accountability, risk management, documentation, and bias mitigation. However, these expectations are expressed at different levels of abstraction and do not necessarily prescribe the technical metrics, evaluation procedures, evidence artefacts, or interpretation criteria required for operational assessment.

Within the scope of Task 2.1, D2.1 addresses three closely related problems. First, relevant Trustworthy AI principles and requirements are distributed across heterogeneous instruments that differ in legal or normative status, terminology, scope, target actors, and level of technical detail, making systematic comparison and consolidation necessary. Second, high-level principles such as Transparency, Fairness, and Accountability must be translated into technical or evidence-based assessment elements that can inform subsequent EXCALIBUR research and implementation activities. Third, the reviewed instruments differ substantially in operational specificity: while some define concrete assessment, documentation, or governance requirements, others remain principle-based and therefore require support from scientific literature to identify suitable metrics, controls, evaluation procedures, and interpretation criteria.

D2.1 addresses these problems through a structured mapping process that connects Trustworthy AI principles to relevant instruments and subsequently to technical metrics, qualitative controls, documentation artefacts, and evidence requirements. This reflects the role assigned to WP2 in the EXCALIBUR work plan: mapping legal and social aspects to technical specifications and providing requirements for subsequent research and technical implementation. D2.1 therefore serves as the initial bridge between the regulatory and conceptual foundations of Trustworthy AI and the operational framework required by later EXCALIBUR activities.

1.4 Regulatory, standardisation, and scientific foundations

Since 2019, a coordinated regulatory, governance, and standardisation response to AI risk has emerged. The EU AI Act [9] establishes binding requirements for AI systems within its scope, including provisions concerning data and data governance, technical documentation, record-keeping, transparency and information provision, and human oversight. Alongside binding regulation, the NIST AI Risk Management Framework [49] provides a non-binding governance structure organised around the functions Govern, Map, Measure, and Manage. The ISO/IEC JTC 1/SC 42 work programme includes standards and technical documents on AI concepts and terminology, bias, trustworthiness, risk management, and AI management systems, including ISO/IEC 23894 [51] and ISO/IEC 42001 [38]. IEEE standards address complementary areas including ethically aligned system design, transparency, and algorithmic bias.

These instruments are important because they translate high-level ethical commitments into more concrete expectations, constrain the design and governance space within which AI systems are increasingly developed and deployed, and establish accountability mechanisms such as documentation, auditability, risk management, conformity assessment, and post-deployment monitoring. For EXCALIBUR, they provide an external reference point: the project is not developing a legal compliance tool, but its technical outputs should remain traceable to documented regulatory, standardisation, and governance expectations rather than relying solely on researcher-defined criteria.

At the same time, the instruments have limitations that are directly relevant to D2.1. Under-specification is common: provisions may identify what should be achieved without prescribing the metric, method, threshold, or evaluation protocol that should demonstrate achievement. Transparency requirements, for example, do not generally prescribe a single XAI method or a universal explanation-quality threshold; similarly, bias-related requirements do not establish one universally applicable fairness metric. The scientific literature contains multiple fairness definitions, bias metrics, explanation methods, and evaluation protocols, some of which encode different assumptions or may be mutually incompatible [21, 23, 25].

Cross-instrument variation is another limitation. The same term may be used differently across regulation, standards, governance frameworks, and scientific literature. Transparency may refer primarily to information disclosure in one instrument, to explainability in another, and to stakeholder-specific information requirements in a third. D2.1 records these differences rather than assuming that similarly named concepts are operationally identical. A further limitation is the pace mismatch between regulation and technology. Regulatory and standardisation cycles are comparatively slow, while AI methods — particularly general-purpose and generative AI — evolve rapidly. This creates areas in which implementation guidance remains incomplete or unsettled. Accountability mechanisms are also heterogeneous: model cards, datasheets, audit logs, risk registers, and related artefacts can provide useful evidence, but their mere presence does not by itself establish completeness, quality, or effectiveness.

These limitations define the analytical role of D2.1. The deliverable identifies which expectations can be operationalised directly, which remain high-level or ambiguous, and where scientific evidence is required to support the selection and interpretation of metrics, controls, and possible thresholds. Scientific literature therefore complements — but does not replace — the regulatory, standardisation, governance, and normative analysis.

1.5 Scope, contribution, and structure of the deliverable

D2.1 is the first deliverable of WP2 and provides the initial mapping foundation for EXCALIBUR's work on Trustworthy AI principles. Its scope follows Task 2.1, which is dedicated to Trustworthy AI principles' mapping, with particular emphasis on Transparency, Fairness, and Accountability and on translating these principles into a technical framework. The deliverable does not propose a new regulatory framework. Instead, it performs a structured and traceable mapping designed to support the needs of Task 2.1 and subsequent EXCALIBUR activities.

While Task 2.1 in the project proposal initially emphasised EU legislative and Trustworthy AI frameworks, the implemented mapping was broadened to selected international standards, governance frameworks, and comparative instruments where these provide complementary terminology, operational guidance, or assessment evidence relevant to the three pillars. This extension preserves the original Task 2.1 objective while strengthening the technical and comparative basis of the mapping.

The analysis combines two evidence streams. The first consists of regulatory, standardisation, governance, and normative instruments relevant to Trustworthy AI. These instruments are organised in an initial landscape table that

records their type, issuing body, legal or normative status, and relevance to the three Task 2.1 pillars. The second consists of the scientific literature corpus collected for Task 2.1, which is used to identify technical metrics, formulas, controls, evaluation procedures, documentation artefacts, and evidence-based interpretation criteria that can support operationalisation.

The contribution of D2.1 is threefold. First, it constructs a broad and structured landscape of relevant instruments and provides a transparent rationale for selecting the instruments carried forward for deeper analysis. Second, it maps selected principles, provisions, requirements, and guidance to measurable or auditable elements, with the resulting metrics-and-thresholds mappings presented in Tables 6–8.

Third, it consolidates these mappings into an EXCALIBUR-oriented taxonomy and requirements structure in which the provenance and interpretation of each operational element can be traced.

The remainder of the deliverable is structured as follows. §2 describes the methodology for instrument identification, scientific-literature selection, extraction, mapping, and threshold or interpretation-criterion classification. §3 defines the operational meaning of the three pillars used throughout the report. §4 presents the landscape of regulations, standards, and frameworks and explains the selection of the core instruments. §5 operationalises the selected instruments through metrics, controls, and evidence artefacts. §6 examines metrics, thresholds, and evidence-based interpretation criteria. §7 consolidates the results into an EXCALIBUR-oriented taxonomy, requirements structure, and indicative scenarios. §8 concludes the deliverable and identifies the implications for subsequent EXCALIBUR work.

2 Methodology

2.1 Methodological overview

The methodology of D2.1 follows a staged mapping process that connects Trustworthy AI principles to relevant instruments and, subsequently, to technical and evidence-based assessment elements. Two parallel streams are used: regulatory, standardisation, governance, and normative instruments on the one hand, and the scientific literature corpus collected for Task 2.1 on the other. The scientific literature is required because many instruments define what should be achieved but do not specify which metric, formula, threshold, or evaluation protocol should be used; its role is therefore to complement, and not to replace, the regulatory and standardisation mapping. The approach is a structured mapping exercise rather than a formal systematic review: the aim is not to cover exhaustively all legal instruments or all scientific publications on trustworthy AI, but to identify and organise those that are most relevant to the Task 2.1 scope, while supporting traceable mappings from instruments and publications to requirements, metrics, controls, and evidence. Additional material may be incorporated in later project activities if new regulatory or technical needs emerge. The five stages of the process are summarised below and illustrated in Figure 1.

1. Instrument inventory: a broad inventory is constructed of regulatory, standardisation, governance, and normative instruments relevant to trustworthy AI.
2. Pillar relevance mapping: each instrument is mapped against the three Task 2.1 pillars, Transparency, Fairness, and Accountability, in order to identify its level and type of relevance.
3. Selection of core instruments: a smaller set of core instruments is selected for deeper analysis on the basis of relevance to Task 2.1, authority of the issuing body, applicability to AI systems, and operational value for EXCALIBUR.
4. Operationalisation and evidence mapping: the selected instruments are connected to technical metrics, qualitative controls, documentation artefacts, and, where possible, thresholds or evidence-based interpretation criteria, drawing on the scientific literature corpus to support the operationalisation of high-level requirements.
5. Consolidation into EXCALIBUR requirements: the mapped elements are consolidated into the EXCALIBUR taxonomy of trustworthy-AI requirements, which is carried forward to WP3, WP4, and WP5.

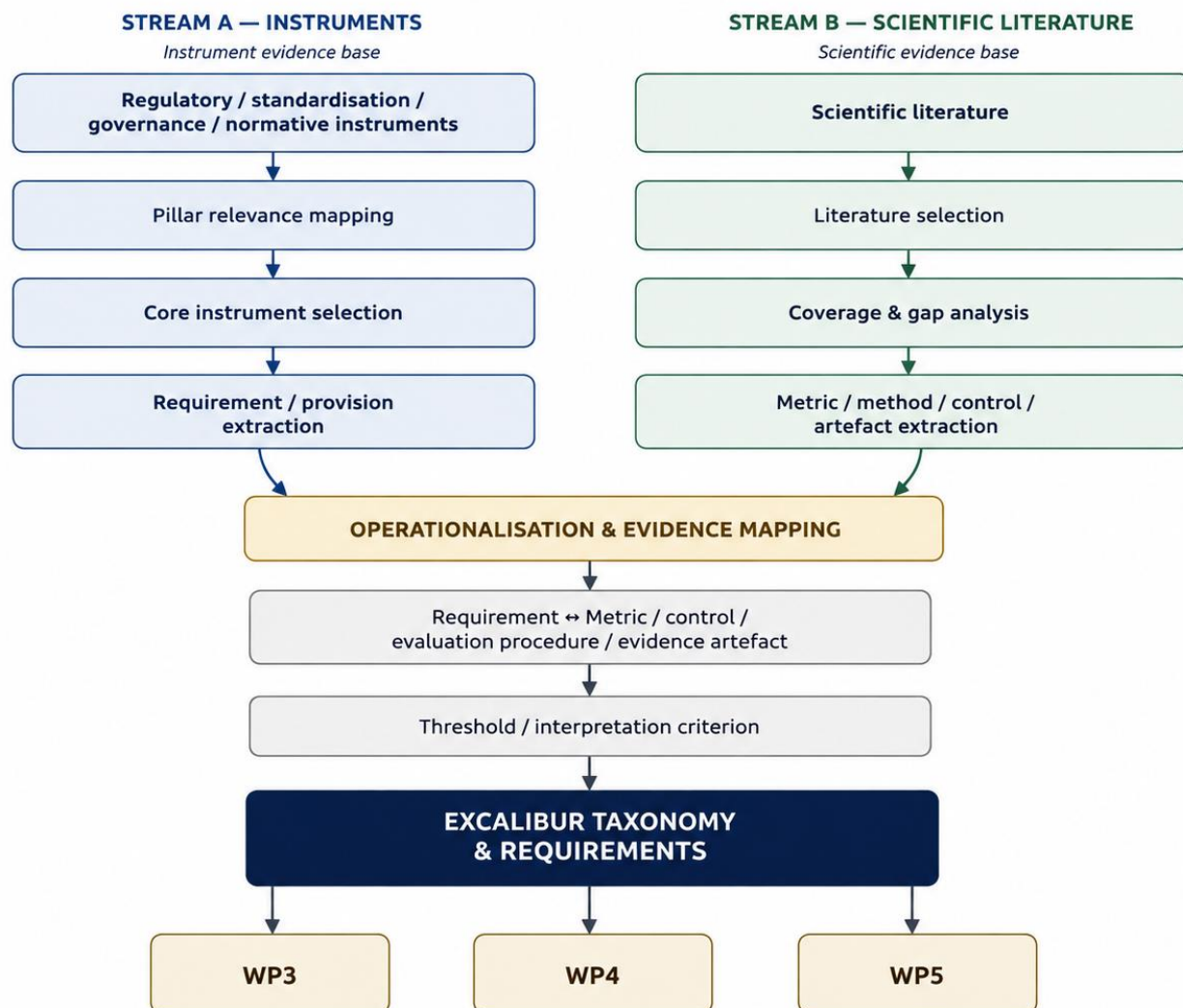


Figure 1. Two-stream methodology of D2.1: instrument analysis (Stream A) and scientific literature analysis (Stream B) converge in the operationalisation and evidence mapping stage, which produces the EXCALIBUR taxonomy and requirements carried forward to WP3, WP4, and WP5.

2.2 Identification of regulatory, standardisation, and normative instruments

2.2.1 Sources and inclusion criteria

Regulatory, standardisation, governance, and normative instruments were identified from the official catalogues, portals, and repositories of the issuing bodies. Relevant instruments were identified through EUR-Lex for European Union legislative instruments; the ISO/IEC and IEEE standards catalogues for international standardisation bodies; national and intergovernmental AI portals such as the NIST AI portal, OECD.AI, UNESCO, the Council of Europe, and U.S. White House resources; and relevant professional and scientific organisations such as ACM. The inclusion criteria were applied as follows:

- The instrument is AI-specific or contains explicit AI-related provisions, concepts, requirements, or assessment criteria.
- The instrument addresses at least one of the three EXCALIBUR pillars — Transparency, Fairness, or Accountability — either through operative provisions, assessment items, technical guidance, or explicit definitions.
- The instrument is published, formally adopted, or available in an advanced and stable version at the time of analysis.

- The instrument is officially published or formally available through the issuing organisation, and sufficient English-language documentation is available to support classification and mapping.

The exclusion criteria were the following:

- Domain-specific regulation touching AI only incidentally, without specifying AI-related requirements or implications.
- Early-stage drafts where the scope, terminology, or obligations remain unstable.
- Voluntary codes from individual companies, unless formally adopted by a recognised standardisation, regulatory, professional, or intergovernmental body.
- Instruments that refer to trustworthy AI only at a very general level without contributing to the three Task 2.1 pillars or to the construction of the EXCALIBUR mapping.

The application of these criteria yielded the initial set of instruments listed in §4.1.

2.2.2 Initial relevance mapping and selection of core instruments

After constructing the initial inventory, each instrument was mapped against Transparency, Fairness, and Accountability. The purpose of this first-level mapping was not to perform a full technical assessment of each instrument, but to identify which instruments are relevant to the EXCALIBUR Task 2.1 objectives and which should be carried forward for deeper analysis.

For each instrument, the mapping records:

- the type of instrument, for example regulation, standard, framework, assessment list, guideline, or recommendation;
- its issuing body and its legal or normative status;
- whether the instrument addresses Transparency, Fairness, and Accountability — and at what level, using the three-level scoring scheme of Table 1 in §4.1: clause-level operative provisions (✓), principle-level or definitional reference only (○), or silent / only incidental (—);
- whether the instrument provides direct operational guidance, general principles, technical terminology, documentation requirements, audit requirements, or only contextual relevance.

The shortlist of core instruments for deeper analysis is selected by combining the per-pillar score above with a set of relevance and authority criteria: relevance to Task 2.1, authority and credibility of the issuing body, applicability to AI systems, operational value for metrics or controls, complementarity with other instruments, and relevance to the EXCALIBUR framework and toolset. This selection step ensures that the deliverable does not treat all reviewed instruments as equally important: it aligns the analysis with the objectives and scope of the Grant Agreement and the tasks defined in Task 2.1, and distinguishes between instruments that shape the core mapping and instruments that provide contextual or supporting material. §4.2 reports the outcome of this selection. Newer instruments such as ISO/IEC 42001:2023 and the U.S. OMB M-26-04 memorandum are recognised as supplementary guidance; they inform our analysis but do not appear in the core shortlist because their specialised scope complements, rather than defines, the five-instrument core mapping adopted for Task 2.1.

2.3 Identification of scientific literature

2.3.1 Sources and inclusion criteria

The scientific literature corpus consists of an initial set of papers collected for Task 2.1 and recorded in the accompanying literature review dataset. The corpus was curated for direct relevance to the EXCALIBUR mapping and is organised around the three pillars of the deliverable: Transparency, Fairness, and Accountability.

The initial corpus was selected according to the following criteria:

- The paper addresses at least one of the three pillars at the level of method, metric, evaluation protocol, taxonomy, or technical framework, rather than only providing general commentary.
- The paper provides formal definitions, formulas, methodological categories, evaluation criteria, or operational concepts that can support the mapping from regulations and standards to technical assessment elements.
- The corpus prioritises peer-reviewed journal and conference papers. Preprints are included only when they are directly relevant to the mapping, provide formal definitions or metrics, and have clear research uptake or conceptual value for EXCALIBUR.
- The corpus mainly focuses on post-2016 work, reflecting the rapid development of fairness, explainability, and AI governance research after the GDPR period. Foundational earlier papers (e.g., Dwork et al. 2012 [10]) are retained where they introduce definitions or metrics that remain central to the field.
- The paper contributes to the operationalisation of at least one Trustworthy AI principle, provision, requirement, or assessment need, for example by defining fairness metrics, explainability evaluation criteria, accountability artefacts, or human-centred evaluation methods.

The literature corpus is used as technical evidence for the metric and threshold mapping. It does not replace the regulatory and standards analysis; rather, it helps translate high-level principles, provisions, requirements, and recommendations into technical or evidence-based assessment elements that can be used within EXCALIBUR. To make the literature mapping traceable, the accompanying Task 2.1 literature dataset records the papers considered, their pillar relevance, the extracted metric or concept, and their contribution to the mapping. The corpus was not designed as a full systematic review; however, it was curated through targeted searches around fairness metrics, explainability/XAI evaluation, accountability artefacts, algorithmic auditing, model documentation, and trustworthy AI governance. This makes the mapping transparent while keeping the deliverable aligned with the practical requirements of Task 2.1.

2.3.2 Coverage and gap analysis

A coverage and gap analysis was conducted on the initial literature corpus to identify which parts of the EXCALIBUR Task 2.1 scope are well covered and which require further support. This analysis is important because the three pillars are not equally mature in the scientific literature. Fairness and explainability have several established metrics and evaluation methods, while accountability is often expressed through documentation, auditability, governance, and process controls rather than through mathematical metrics.

The initial corpus is structured into three broad groups:

- Fairness-oriented literature, including works on bias definitions, fairness taxonomies, group fairness, individual fairness, subgroup fairness, fairness in regression, and trade-offs between competing fairness criteria.
- Explainability-oriented literature, including works on post-hoc explanations, concept-based explanations, explanation faithfulness, robustness, sensitivity, sanity checks, and human-centred evaluation of explanations.

- Accountability and governance literature, including dedicated audit-framework, documentation, and process-metric sources ([32] — SMACTR end-to-end internal audit framework; [16] — Datasheets for Datasets; [15] — Model Cards for Model Reporting; [33] — Responsible AI Metrics Catalogue; [34] — three-layered LLM audit blueprint), together with the accountability-related contributions of the fairness and XAI sub-corpora (algorithmic auditing of subgroup fairness in [14]; algorithmic XAI audit instruments in [28], [30], and [31]; independent human audit protocols in [29]). Accountability is substantively represented in the corpus through dedicated sources on internal auditing, dataset and model documentation, responsible-AI process metrics, and LLM auditing [15, 16, 32-34].

The gap analysis also identifies areas where additional sources may be needed. The remaining gaps concern time-series and wearable data, multimodal settings, LLM-based explainability and fairness, federated and decentralised settings, non-expert human-centred evaluation, and combined XAI–fairness assessment. The remaining gaps are documented explicitly and treated not as weaknesses of the deliverable but as areas for future work that will inform WP3, WP4, and WP5, keeping the mapping transparent and extensible as the EXCALIBUR framework develops.

2.4 Extraction and mapping approach

A common extraction template was applied to both the instrument stream and the scientific literature stream. For each instrument or scientific publication, the following fields were recorded:

- **Pillar:** Transparency, Fairness, Accountability, or cross-cutting.
- **Sub-dimension:** the specific aspect addressed by the item, such as model explainability, disclosure, documentation, group fairness, subgroup fairness, bias audit, risk management, audit trail, or responsibility allocation.
- **Item type:** regulation, standard, technical report, framework, assessment list, recommendation, guidance document, or scientific publication.
- **Target actor or audience:** provider, deployer, developer, auditor, regulator, end-user, affected individual, researcher, or organisation.
- **Operational form:** metric, formula, threshold, qualitative criterion, documentation artefact, audit requirement, checklist item, risk control, or no direct operational element.
- **Evidence basis:** whether the element is grounded in an external instrument, a scientific publication, recognised guidance, or a project-defined interpretation.
- **Relevance to EXCALIBUR:** whether the element can support the EXCALIBUR framework, the toolset, the validation activities, or the taxonomy and requirements structure.

Extraction and classification were performed manually by the Task 2.1 research team using the common template described above. Before consolidation into the final mapping, the extracted entries were cross-checked by the same team against the corresponding instruments and scientific publications to ensure consistency of classification, terminology, and traceability. The extraction focused on retaining traceability between each instrument or scientific publication, the pillar it supports, the type of provision, requirement, or assessment element it expresses, and the metric, control, or evidence artefact that may operationalise it. Where an instrument or publication remained too abstract to support a metric directly, this was recorded explicitly rather than replaced by an unsupported technical assumption.

The mapping approach proceeds in three steps. First, each regulatory, standardisation, governance, or normative instrument is classified according to its relevance to Transparency, Fairness, and Accountability. This produces the broad instrument landscape mapping presented in §4. Second, the most relevant instruments are selected for deeper analysis. For these instruments, the deliverable identifies clauses, requirements, assessment items, or technical guidance that can be connected to measurable or auditable elements. This produces the instrument-to-

metrics and instrument-to-controls mapping presented in §5. Third, the mapped metrics and controls are examined in relation to thresholds or interpretation criteria. This produces the metrics-and-thresholds mapping in §6.

Thresholds are classified into four categories:

- source-defined thresholds, where a regulation, standard, or official framework explicitly provides a threshold or decision rule;
- literature-supported thresholds, where a threshold is proposed, used, or justified in the scientific literature;
- project-defined operational thresholds, where EXCALIBUR may define a practical interpretation rule for experimentation or validation purposes;
- no established threshold, where the metric or control can be reported but should not be judged against a universal cut-off.

The label “source-defined” is used in this deliverable only as a threshold-provenance category. Elsewhere, the deliverable distinguishes between instruments — regulations, standards, and frameworks — and scientific publications, and does not use “source” as a generic label for both.

The final mapping logic follows the sequence: Instrument or scientific publication → Pillar → Principle / provision / requirement → Metric / control / evaluation procedure / evidence requirement → Threshold or interpretation criterion → EXCALIBUR requirement. This traceability structure ensures that the deliverable separates what is explicitly required by regulations or standards, what is supported by scientific literature, what can be operationalised through project-specific decisions, and what remains an open gap for later EXCALIBUR work, thereby avoiding unsupported technical claims.

3 Transparency, Fairness, and Accountability: Operational Definitions and Inter-Pillar Tensions

The mapping developed in this deliverable requires clear operational definitions of Transparency, Fairness, and Accountability. These concepts are used differently across regulations, standards, governance frameworks, and scientific literature. Section 3 therefore specifies how each pillar is interpreted for the purposes of D2.1, including instrument and publication classification, metric/control mapping, and construction of the EXCALIBUR taxonomy. These operational definitions are not intended to establish new universal or normative definitions of the three concepts; rather, they provide a consistent interpretation for the mapping and technical operationalisation carried out in this deliverable.

3.1 Transparency — including Explainability/XAI

Transparency is used in D2.1 as the broader regulatory and governance concept that enables relevant stakeholders to understand, inspect, and question an AI system and the process through which its outputs are produced. Within this broader pillar, Explainability/XAI is treated as the technical operationalisation of model-side transparency, focusing on methods and evaluation criteria that make model behaviour and individual outputs understandable to relevant stakeholders.

The pillar is therefore divided into two complementary dimensions. The model-side dimension covers interpretability and explainability. Interpretability refers to cases where the model — or part of its reasoning — can be directly understood, while explainability refers to post-hoc methods that generate explanations of model behaviour or individual predictions. Representative methods include LIME [11], SHAP [12], concept activation vectors [26], counterfactual explanations [19], and related XAI approaches surveyed in [5]. Their evaluation may involve criteria such as fidelity, faithfulness, infidelity, sensitivity, robustness, sanity checks, and human-centred

usefulness, catalogued by the Quantus framework [28] and individual works [27, 29, 30, 31]. Recent scientific literature motivates training-data attribution and influence-based analysis as optional data-level explanation evidence [44], while supplementary governance and procurement sources emphasise documentation, audit trails, and feedback mechanisms [32–34, 40]. In D2.1, these elements support the technical operationalisation layer and do not constitute additional obligations imposed by the five shortlisted instruments.

The process-side dimension covers disclosure, documentation, traceability, and communication. It concerns whether the use of an AI system is disclosed, whether technical and organisational documentation is available, whether design and deployment decisions can be reconstructed, and whether relevant users or deployers receive sufficient information to understand the system’s capabilities and limitations. In the regulatory landscape, the AI Act addresses the process side through Arts. 11 (technical documentation), 12 (record-keeping), and 13 (transparency and information provision to deployers); IEEE 7001-2021 addresses it through stakeholder-specific transparency levels; GDPR Arts. 13–15 address related information and access rights, including meaningful information about the logic involved in automated decision-making where applicable.

EXCALIBUR is particularly concerned with the model-side dimension because subsequent WP3 work is intended to investigate LLM-based methods for synthesising and communicating explanations. Process-side transparency nonetheless remains essential for the deliverable, because the generated explanations become meaningful only when they are connected to, and interpreted against, documented data, model, evaluation, and deployment choices.

3.2 Fairness — operational definition

Fairness refers to the identification, assessment, and mitigation of systematic disadvantage in AI system outputs, errors, or impacts across individuals, groups, or subgroups defined by protected or sensitive attributes. In D2.1, fairness is treated as a context-dependent technical and governance requirement rather than as a single universal metric. The appropriate fairness criterion depends on the task, the affected population, the protected attributes, the decision context, and the type of harm that the assessment aims to detect. The independence-, separation- and sufficiency-based metrics used in this deliverable rely on classification quantities such as the true positive rate (TPR), false positive rate (FPR), positive predictive value (PPV), negative predictive value (NPV), and group-specific outcome prevalence. These quantities are treated as diagnostic inputs rather than universal fairness thresholds; they underpin the fairness metrics presented in § 6.

Three sub-dimensions structure the mapping [6, 20, 24]. Group fairness evaluates whether outcomes or error rates differ across protected groups. It may be operationalised through metrics such as demographic parity, statistical parity difference, disparate impact ratio, equal opportunity, equalised odds, predictive parity, or calibration. Individual fairness requires that similar individuals receive similar outcomes [10], where similarity must be defined through a task-relevant distance or counterfactual relation [13]. Subgroup fairness extends group fairness to intersectional or overlapping protected categories, addressing cases where a model appears fair on each marginal group but discriminates against intersectional subgroups — the failure mode known as fairness gerrymandering [14].

The worldview commitment behind metric choice is recorded explicitly because it informs which metric families are appropriate and how their results should be interpreted [23]:

- We’re All Equal (WAE) assumes that groups are equal with respect to the task-relevant construct and interprets observed group disparities as potentially reflecting structural bias. Independence-based group-parity metrics, such as demographic parity, statistical parity difference, and disparate impact ratio, are compatible with this worldview.

- What You See Is What You Get (WYSIWYG) assumes that observed features and labels adequately reflect task-relevant constructs. Metrics that rely on the observed target as meaningful — such as equalised odds, equal opportunity, and predictive parity — are compatible with this worldview.

The two worldviews cannot be reconciled by a single metric. Impossibility results established by Chouldechova (2017) [17] and Kleinberg, Mullainathan and Raghavan (2017) [18], and surveyed in [21, 23, 25], show that when base rates differ across groups, calibration or predictive parity generally cannot be satisfied simultaneously with equalised error-rate conditions, except in special cases such as perfect prediction. More broadly, several fairness criteria are mutually incompatible, so metric selection must be justified by the application context. Any fairness mapping in §6 must therefore specify the relevant worldview, metric family, assumptions, and limitations rather than assuming a single “correct” notion.

3.3 Accountability — operational definition

Accountability refers to the capacity to document, justify, audit, and assign responsibility for the design, deployment, monitoring, and outcomes of an AI system and — correspondingly — to the ability of relevant actors to obtain appropriate information, challenge decisions, and seek redress where applicable. In D2.1, accountability is not treated primarily as a mathematical metric. It is operationalised mainly through controls, documentation artefacts, audit trails, governance procedures, and evidence requirements that allow a third party to understand whether transparency and fairness obligations have been considered and acted upon.

Three sub-dimensions structure the pillar. Auditability requires records sufficient for a third party to reconstruct what was decided, on what data, by which model version, under which configuration, and with which evaluation results. The AI Act operationalises this through Art. 11 (technical documentation) and Art. 12 (automatic log generation); the NIST AI RMF locates it in the Govern function; ISO/IEC 42001 codifies it as a management-system requirement. Responsibility allocation requires that obligations be unambiguously assigned along the AI value chain — provider, deployer, distributor, importer — so that no failure mode falls into a responsibility vacuum. The AI Act’s Arts. 16–27 are the most developed treatment. Documentation artefacts refer to structured outputs such as model cards [15], datasheets for datasets [16], the AI Act’s Annex IV technical-documentation requirements, risk registers, impact assessments, monitoring records, bias audit reports, and audit-framework artefacts such as the SMACTR Audit Summary Report [32].

Two qualifications apply. First, model cards and datasheets, while increasingly cited in standards and policy documents, have not yet been standardised: there is no widely accepted quality criterion for a model card, no widely accepted certification scheme for datasheets, and no commonly adopted machine-readable schema for either. §5.3 records this as a coverage gap. Second, accountability is substantively represented in the scientific literature corpus through dedicated work on internal algorithmic auditing [32], dataset and model documentation [15, 16], responsible-AI process metrics [33], and LLM auditing [34]. These contributions are integrated into the accountability mappings in §5.3, §6.4, and §7.2.

Compared with fairness and explainability, accountability is less often expressed through computable metrics. Accordingly, the accountability mapping in D2.1 relies mainly on qualitative and process-oriented controls that assess not only whether relevant artefacts and governance mechanisms exist, but also whether they are sufficiently complete, current, traceable, and suitable for the intended purpose. Relevant evidence may include model cards, audit logs, risk assessments, responsibility assignments, and post-deployment monitoring records. Presence is therefore treated as a necessary but not sufficient indicator of accountability; where feasible, the quality and completeness of the supporting evidence should also be assessed. As a supplementary non-European example, the U.S. OMB M-26-04 memorandum shows how LLM accountability evidence can become part of public-sector procurement. It requires U.S. federal agencies procuring LLMs to request vendor documentation such as acceptable-use policies, model/system/data cards, end-user resources, and feedback mechanisms, while additional model-

evaluation evidence may be requested depending on the intended use. In D2.1, this source is treated only as an illustrative procurement reference, not as a core EXCALIBUR requirement.

3.4 Inter-pillar tensions

The three pillars cannot always be optimised independently. Their implementation may create tensions that must be recorded during the mapping and later addressed during the design of the EXCALIBUR framework.

Fairness and predictive performance. Introducing fairness constraints or mitigation procedures may affect aggregate predictive performance, especially when the selected fairness criterion changes the decision boundary, reweights training data, or adjusts model outputs [22, 25]. This does not mean that fairness necessarily reduces performance in every setting, but it means that the trade-off must be measured and documented.

Transparency and privacy. Detailed explanations and other forms of model output or disclosure may increase information exposure concerning sensitive features, model behaviour, or training data, while machine-learning models can also be vulnerable to model-inversion and membership-inference attacks [52, 53]. This creates a tension between explanation richness and privacy protection, especially in human-centric domains such as wearables and health-related inference.

Fairness and data minimisation. Several group-fairness metrics require access to protected or sensitive attributes. Some may constitute special categories of personal data under GDPR Art. 9, while others may be protected under non-discrimination law without falling within Art. 9. Where personal data are processed, an appropriate legal basis, data minimisation, and the applicable safeguards are required. This creates a practical tension between measuring bias and minimising and justifying personal-data processing.

Accountability and development agility. Detailed documentation, audit trails, and monitoring procedures improve accountability, but they may also slow down model iteration and deployment. For EXCALIBUR, this tension should be addressed by designing lightweight but traceable artefacts that support auditability without making the framework impractical.

Explanation faithfulness and accessibility for non-experts. This tension is intra-pillar within Transparency, but central to the EXCALIBUR methodological hypothesis. Technical explanations — including SHAP values, concept attribution scores, saliency maps, attention-based explanations, or other attribution outputs — aim to represent aspects of the underlying model’s behaviour, but their faithfulness cannot be assumed by construction. Their reliability must be assessed through fidelity, robustness, sensitivity, sanity-check, or human-centred evaluation criteria. Accessible explanations, particularly natural-language narratives generated by LLMs from such technical outputs, may improve usability for non-experts, but they introduce an additional translation layer that must be evaluated for faithfulness, over-simplification, and possible hallucinated reasoning. The EXCALIBUR framework operates inside this trade-off rather than resolving it: §6 therefore distinguishes faithfulness metrics (the model-side property of an explanation) from accessibility / human-centred evaluation criteria [29] and records both for each LLM-generated explanation.

D2.1 does not propose a universal resolution of these tensions. Instead, it identifies them as context-dependent considerations for the later metrics, controls, and requirements mapping. Where a material trade-off is identified for a specific audit, explanation, or use case, the EXCALIBUR framework should record the trade-off, its rationale, and the supporting evidence rather than assuming that all objectives can always be optimised simultaneously.

4 The Landscape of Regulations, Standards and Frameworks

This section presents the first main output of D2.1: a broad landscape table of regulatory, standardisation, governance, and normative instruments relevant to Trustworthy AI. The purpose of this landscape is not to treat all instruments as equally important, but to provide a transparent basis for selecting those analysed in more detail in §5

and §6. The analysis therefore proceeds in two steps. First, §4.1 presents the initial instrument inventory and maps each instrument against Fairness, Transparency (including Explainability/XAI), and Accountability. Second, §4.2 explains which instruments are selected as core instruments for EXCALIBUR and why.

4.1 Instruments examined

Table 1 presents the broad landscape of regulatory, standardisation, governance, and normative instruments examined in EXCALIBUR and maps each instrument against the three Task 2.1 pillars: Fairness, Transparency/XAI, and Accountability. The table uses a three-level coverage indicator to distinguish clause-level operative provisions, principle-level or definitional references, and cases where a pillar is silent or only incidental.

Among the four binding instruments currently in force in the landscape — the EU AI Act, GDPR, Digital Services Act, and NYC Local Law 144 — Transparency and Accountability are represented at clause level across all four, whereas Fairness is represented at clause level in two of the four: the EU AI Act and NYC Local Law 144. GDPR and the Digital Services Act address fairness only at principle or indirect level under the scoring scheme used in this deliverable. The Council of Europe Framework Convention on Artificial Intelligence is retained in the landscape as an important international instrument, but it is not counted among the binding instruments currently in force because the Convention has not yet entered into force.

The regulatory landscape has evolved significantly during the preparation of the deliverable. The AI Act entered into force on 1 August 2024, with phased application of its provisions. Regulation (EU) 2026/1744, the Digital Omnibus on AI, entered into force on 27 July 2026 and amended the AI Act implementation timeline, including application from 2 December 2027 for high-risk AI systems classified under Article 6(2) and Annex III and from 2 August 2028 for high-risk AI systems under Article 6(1) and Annex I [35]. Article 50 transparency obligations apply from 2 August 2026. In July 2026, the European Commission also published Guidelines on the implementation of the Article 50 transparency obligations, complementing the Code of Practice on Transparency of AI-generated Content published in June 2026 [54, 55].

Outside the EU, the NIST AI Risk Management Framework continues to develop; it has published a Generative AI Profile (July 2024) [37] and a concept note for a Critical Infrastructure profile (April 2026) [47], while ISO/IEC 42001:2023 provides a structured AI governance framework [38] complementing ISO/IEC 23894. Although the deliverable retains ISO/IEC 23894 as the primary risk-management standard, we acknowledge that ISO/IEC 42001 will play an important role in shaping AI governance systems. A 2025 New York State Comptroller audit found weaknesses in DCWP’s enforcement of Local Law 144 and recommended stronger complaint handling, more complete bias-audit review, proactive identification of potential non-compliance, and greater use of technical expertise [48]. The law continues to impose three main obligations: an independent bias audit conducted within the preceding year, with calculation of selection rates or scoring rates, as applicable, and the corresponding impact ratios; publication of a summary of the audit results; and notice to covered candidates or employees at least ten business days before use of the AEDT, including the prescribed information [39].

#	Instrument	Type	Issuer / Jurisdiction	Status	F	T/XAI	A
1	EU AI Act (Reg. 2024/1689)	Regulation	EU	Binding EU regulation · in force 1 Aug 2024 · phased application 2025–2028	✓	✓	✓
2	GDPR (Reg. 2016/679)	Regulation	EU	Binding · in force 2018	○	✓	✓
3	Digital Services Act (Reg. 2022/2065)	Regulation	EU	Binding · fully applicable 2024	○	✓	✓
4	CoE Framework Convention on AI (CETS 225)	Int. Convention	Council of Europe	Not yet in force · opened for signature 5 Sep 2024	✓	✓	✓
5	NYC Local Law 144 of 2021 (AEDT Law)	Local law	NYC (US municipal)	Binding · effective 2023, enforced Jul 2023	✓	✓	✓
6	AI-HLEG Ethics Guidelines for Trustworthy AI	Guidelines	EU (HLEG)	Voluntary · 2019	○	○	○
7	ALTAI — Assessment List for Trustworthy AI	Assessment list	EU (HLEG)	Voluntary · 2020	✓	✓	✓
8	OECD AI Principles	Principles	OECD	Voluntary · 2019, rev. 2024	○	○	○
9	UNESCO Recommendation on the Ethics of AI	Recommendation	UNESCO	Voluntary · 2021	○	○	○
10	NIST AI Risk Management Framework 1.0	Framework	NIST (US federal)	Voluntary · 2023	✓	✓	✓
11	Blueprint for an AI Bill of Rights	Policy blueprint	OSTP (US federal)	Voluntary · 2022	✓	✓	○
12	ISO/IEC 22989:2022 — AI concepts and terminology	Int. standard	ISO/IEC JTC 1/SC 42	Voluntary · 2022	○	○	○
13	ISO/IEC 23053:2022 — Framework for AI using ML	Int. standard	ISO/IEC JTC 1/SC 42	Voluntary · 2022	○	○	○
14	ISO/IEC 23894:2023 — AI risk management	Int. standard	ISO/IEC JTC 1/SC 42	Voluntary · 2023	✓	✓	✓
15	ISO/IEC TR 24027:2021 — Bias in AI systems and AI aided decision making	Technical report	ISO/IEC JTC 1/SC 42	Voluntary · 2021	✓	—	○
16	ISO/IEC TR 24028:2020 — Trustworthiness in AI	Technical report	ISO/IEC JTC 1/SC 42	Voluntary · 2020	○	○	○
17	ISO/IEC 42001:2023 — AI management system	Int. standard	ISO/IEC JTC 1/SC 42	Voluntary (certifiable) · 2023	○	○	✓
18	ISO/IEC TS 12791:2024 — Treatment of unwanted bias in ML	Technical spec.	ISO/IEC JTC 1/SC 42	Voluntary · 2024	✓	—	—
19	IEEE 7000-2021 — Ethical concerns in system design	Standard	IEEE	Voluntary · 2021	○	○	○
20	IEEE 7001-2021 — Transparency of autonomous systems	Standard	IEEE	Voluntary · 2021	—	✓	○
21	IEEE 7003-2024 — Algorithmic bias considerations	Standard	IEEE	Voluntary · 2024/2025	✓	—	○

Table 1 — Regulations, standards, frameworks, and guidelines relevant to trustworthy AI.

Legend. ✓ = clause-level operative provisions (named obligations, specific procedural or disclosure requirements, named metrics or artefacts); ○ = principle-level or definitional reference only; — = silent or only incidental. F = Fairness, T/XAI = Transparency, including Explainability/XAI where the source addresses model-side explanation or interpretability, A = Accountability. Row shading is used only as a visual aid to distinguish instruments and broad source families. The colours do not encode legal status, pillar coverage, or shortlist membership; these are indicated by the Status column, the ✓/○/— symbols, and Table 2, respectively.

4.2 Selection rationale: shortlist of five instruments

The selection of five core instruments is based on per-pillar coverage, the authority of the issuing body, legal force (binding vs voluntary), practical value for EXCALIBUR and complementarity. We keep the original shortlist (EU AI Act, NIST AI RMF, ISO/IEC 23894, NYC Local Law 144, OECD AI Principles) but recognise several important updates:

The AI Act remains the primary binding framework. The GPAI Code of Practice and the public training-content summary template provide implementation support for the transparency, copyright, and documentation obligations applicable to general-purpose AI model providers [36]. Regulation (EU) 2026/1744 entered into force on 27 July 2026 and revised the application timeline for high-risk AI-system requirements, including application from 2 December 2027 for systems classified as high-risk under Article 6(2) and Annex III and from 2 August 2028 for systems classified as high-risk under Article 6(1) and Annex I [35]. Article 50 transparency obligations apply from 2 August 2026 and are supported by the Commission Guidelines published on 20 July 2026 and the Code of Practice on Transparency of AI-generated Content published on 10 June 2026 [54, 55].

The NIST AI RMF is still voluntary but has been enriched with the Generative AI Profile [37] and the plan for a Critical Infrastructure profile [47], signalling increased sector-specific guidance.

ISO/IEC 42001 now provides a standardised AI management system and can complement ISO/IEC 23894. We consider this framework in the interactions with risk management without altering the core shortlist [38].

The significance of NYC Local Law 144 has increased due to the annual bias audit requirement and the recent push for stronger enforcement [39, 48].

The OECD AI Principles remain in the list because they supply high-level values and terminology, despite lacking specific obligations.

The selected instruments continue to offer the optimal mix of binding force and practical value for setting EXCALIBUR requirements. The OECD AI Principles are retained not for clause-level coverage but as the overarching normative framework that ties the shortlist together. As the first intergovernmental AI standard (2019, updated 2024) and the basis for the G20 AI Principles, they provide the shared definitions of Transparency, Fairness and Accountability on which the operative instruments draw, letting the AI Act, NYC Local Law 144, the NIST AI RMF and ISO/IEC 23894 be read as instances of a common framework rather than isolated requirements. Their principle-level scoring (o) reflects this integrative role and is intentional, not a coverage gap. Instruments such as ALTAI and ISO/IEC 42001 remain important supporting references. ALTAI informs the broader trustworthy-AI assessment logic, while ISO/IEC 42001 complements the governance and management-system perspective. They are not included in the five-instrument core shortlist because D2.1 prioritises sources that jointly provide legal force, international risk-management structure, operational audit comparators, and principle-level framing for the three Task 2.1 pillars.

Table 2 summarises the five instruments selected for deeper operational analysis and clarifies the distinct role that each one plays in the EXCALIBUR mapping. The shortlist combines the binding and technically broad EU AI Act, the operational fairness-audit comparator provided by NYC Local Law 144, the risk-management and governance perspectives of the NIST AI RMF and ISO/IEC 23894, and the principle-level framing of the OECD AI Principles. The table therefore makes explicit that the five instruments are complementary rather than equivalent in legal force or operational specificity.

#	Instrument	Status	F	T/XAI	A	Primary contribution to EXCALIBUR mapping
S1	EU AI Act (Reg. 2024/1689)	Binding EU regulation · in force 1 Aug 2024 · phased application 2025–2028	✓	✓	✓	Horizontal AI regulation; high-risk obligations across data, documentation, transparency, oversight.
S2	NYC Local Law 144 of 2021	Binding (local) · 2023	✓	✓	✓	Operational fairness-audit comparator retained because the DCWP Final Rule provides source-defined calculation rules and worked examples for selection rates or scoring rates, as applicable, and the corresponding impact ratios across protected and intersectional categories, together with public-disclosure and notice requirements [39].
S3	NIST AI RMF 1.0	Voluntary · 2023	✓	✓	✓	Widely used internationally; named sub-functions per pillar (Measure 2.11, Measure 2.9, Govern 2.1).
S4	OECD AI Principles	Voluntary · 2019, rev. 2024	○	○	○	Principle-level international reference point; values baseline for fairness, transparency, and accountability without operative clauses.
S5	ISO/IEC 23894:2023	Voluntary · 2023	✓	✓	✓	Relevant AI risk management standard supporting AI Act-oriented risk management, documentation, and control practices.

Table 2 — Shortlist of instruments carried forward to §5 and §6.

5 Operationalising the Selected Instruments

This section links the five selected instruments to measurable elements and controls. Because the shortlisted instruments differ in legal and normative status and most do not prescribe specific metrics or thresholds, the mapping draws on scientific literature to connect provisions, requirements, and guidance to practical assessment elements. The mapping is many-to-many: a provision or guidance element may correspond to several technical methods, while a metric or control may support several instruments. The results are organised into three subsections — Fairness, Transparency/XAI, and Accountability — that mirror the three D2.1 pillars.

During the preparation of D2.1, several relevant regulatory developments became applicable or were finalised. The AI Act obligations for general-purpose AI models have applied since 2 August 2025 and are supported by the GPAI Code of Practice and the public training-content summary template [36]. Regulation (EU) 2026/1744 entered into force on 27 July 2026 and revised the application timeline for high-risk AI-system requirements to 2 December 2027 for systems classified under Article 6(2) and Annex III and 2 August 2028 for systems classified under Article 6(1) and Annex I [35]. Article 50 transparency obligations have applied since 2 August 2026 and are complemented by the Commission Guidelines and the Code of Practice on Transparency of AI-generated Content [54, 55].

5.1 Fairness coverage and operational mapping across the shortlist

In the AI Act, fairness-related obligations arise mainly from Article 10. For high-risk AI systems that use techniques involving the training of AI models with data, the relevant training, validation and testing datasets must be subject to appropriate data-governance practices; they must be relevant, sufficiently representative and, to the best extent possible, free of errors and complete, and they must be examined for possible biases likely to affect health and safety, fundamental rights, or discrimination prohibited under Union law [9]. Following Regulation (EU) 2026/1744, Article 10(5) was deleted and the exceptional processing of special categories of personal data for bias detection and correction is addressed under the new Article 4a, subject to the conditions and safeguards specified therein [35]. However, the AI Act does not define specific Fairness metrics—the choice of metric (e.g., demographic parity, equalised odds, predictive parity, calibration) remains context-dependent and must be justified with reference to the literature. New fairness metrics such as UCerF [41] incorporate model confidence and provide finer-grained assessments of bias. However, these metrics remain experimental and should complement rather than replace established fairness definitions.

New York City Local Law 144 and the DCWP Final Rule require an independent bias audit conducted within the preceding year. Depending on the AEDT output, the audit must calculate selection rates or scoring rates, as applicable, and the corresponding impact ratios across sex, race/ethnicity, and intersectional categories; the Final Rule also provides worked examples of these calculations [39]. Employers and employment agencies must publish a summary of the audit results and provide the prescribed notice before using the AEDT [39].

The NIST AI RMF [49] encourages the measurement and management of algorithmic biases, but neither the Generative AI Profile [37] nor the Critical Infrastructure concept note [47] proposes specific Fairness metrics. ISO/IEC 23894 provides guidance on AI risk management, while ISO/IEC 42001 specifies requirements for an AI management system. Both can support the management of bias and fairness risks, but neither prescribes fairness metrics or numerical thresholds [38, 51]. The OECD AI Principles remain high-level [50]. For the operational mapping, fairness-related elements are organised into (a) data examination and documentation, (b) outcome evaluation, and (c) bias mitigation. Where an instrument does not prescribe mitigation methods, candidate methods are drawn from the scientific literature (e.g., reweighting, post-processing, counterfactual fairness).

Table 3 consolidates the fairness-related provisions and guidance of the five shortlisted instruments and links them to candidate technical metrics and methods. The comparison highlights an important difference in operational specificity: the EU AI Act establishes binding data-governance and bias-examination requirements without prescribing a specific fairness metric, while NYC Local Law 144 and the DCWP Final Rule provide the most explicit calculation and reporting procedures through selection rates, scoring rates, and impact ratios. The NIST AI RMF, ISO/IEC 23894, and OECD AI Principles provide broader measurement, risk-management, or principle-level guidance that requires further technical operationalisation.

Instrument	Key clause(s)	What must be evaluated / produced	Candidate metrics / methods	Sources
EU AI Act (Reg. 2024/1689)	Art. 10(2)(f–g), Art. 10(3), Art. 4a	Training data examined for biases affecting protected groups; data representativity; where special-category data are processed for bias detection or correction, necessity and safeguards under Article 4a must be documented	Statistical parity difference; disparate impact ratio; equalised odds; equal opportunity; subgroup analysis per intersectional category	[6, 10, 14, 20, 21, 23, 24]
NYC Local Law 144 of 2021	§20-871; Rules §5-301	Selection rate or scoring rate per required category; impact ratio per category; separate sex, race/ethnicity, and intersectional analysis; number of individuals assigned to unknown categories.	Selection rate = selected individuals / total individuals in the category. Scoring rate = proportion of individuals in the category receiving a score above the sample median. Impact ratio = category selection or scoring rate / highest corresponding category rate. The DCWP Final Rule provides worked calculation examples [39].	NYC LL 144 and DCWP Final Rule [39]
NIST AI RMF 1.0	Map 5.1; Measure 2.11; Manage function (risk treatment)	Documented impact analysis on groups; bias evaluation results; risk treatments selected and applied	Standard group-fairness metrics; subgroup analysis; bias-treatment techniques from NIST AI RMF Playbook	[14, 20, 22, 24]
OECD AI Principles	Principle 1.2 — respect for the rule of law, human rights and democratic values, including fairness and privacy	Principle-level commitment to fairness and non-discrimination; no operative metric, procedure, or threshold specified	None specified; provides values-level orientation to be operationalised through other sources	OECD AI Principles (2019, rev. 2024)
ISO/IEC 23894:2023	Risk identification, analysis, and treatment clauses (AI-scoped from ISO 31000)	Bias risk register entries; treatment plans documented; residual risk after mitigation	Methods from ISO/IEC TR 24027 (bias taxonomy) and ISO/IEC TS 12791:2024 (mitigation techniques); fair regression [22]	ISO/IEC TR 24027; ISO/IEC TS 12791:2024; [22]

Table 3 — Fairness: operative clauses and candidate metrics.

Legend: instruments marked ✓ in the F column of Table 1 carry operative fairness provisions; the OECD AI Principles are included as a principle-level reference. Numbered references correspond to the bibliography; formal fairness metric definitions are provided in Appendix A.

5.2 Transparency/XAI coverage and operational mapping across the shortlist

Transparency obligations address both model-side explainability and process-side disclosure and documentation. Article 11 requires the technical documentation of a high-risk AI system to be drawn up before it is placed on the market or put into service and kept up to date, while Article 12 requires the system to technically allow the automatic recording of events (logs) over its lifetime. Article 13 requires information and instructions for deployers, while Article 14 addresses human oversight. Article 50 introduces transparency obligations for providers and deployers, including notification when interacting with AI systems and disclosure or marking duties for AI-generated or manipulated content. The obligations for general-purpose AI models apply from 2 August 2025; the voluntary GPAI Code of Practice and the public training-data summary template published in July 2025 provide implementation tools for model transparency and documentation [36]. Article 50 transparency obligations apply from 2 August 2026 [9]. The Commission Guidelines published on 20 July 2026 clarify the implementation and scope of these obligations, while the Code of Practice on Transparency of AI-generated Content supports implementation of the marking and labelling requirements for AI-generated or manipulated content [54, 55].

In NYC Local Law 144, transparency focuses on publishing the bias-audit summary and providing timely notice to candidates [39]. The NIST AI RMF addresses both procedural transparency and model explanation: MEASURE 2.8 concerns transparency-related risks, while MEASURE 2.9 addresses whether models are explained, validated, and documented, without prescribing a specific XAI method or numerical explanation-quality threshold. The Generative AI Profile recommends documenting training-data provenance and curation policies, monitoring generated outputs and system behaviour, and recording incident and user-feedback information, without mandating specific explanation methods [37]. The OECD AI Principles highlight transparency but do not guide the form of explanations. Scientific literature motivates training-data attribution and influence-based analysis as optional data-level explanation evidence [44], while audit trails and contestability mechanisms are treated as accountability controls. These elements may strengthen traceability, but they do not constitute additional obligations imposed by the five shortlisted instruments.

The key observation remains: the NIST AI RMF explicitly addresses model explanation in MEASURE 2.9. The AI Act requires sufficient transparency and information enabling deployers to interpret and appropriately use the outputs of high-risk AI systems under Article 13, but it does not prescribe a specific XAI method, explanation format, or numerical explanation-quality threshold. The OECD AI Principles recognise explainability at principle level; NYC Local Law 144 focuses on notice and public bias-audit disclosure; and ISO/IEC 23894 provides risk-management guidance. None of the five shortlisted instruments prescribes a universal XAI method or explanation-quality threshold [9, 39, 49–51].

Table 4 summarises how the five shortlisted instruments address Transparency/XAI and connects their provisions or principles to candidate methods and evidence. A central finding is that the shortlist is considerably stronger on process-side transparency — documentation, disclosure, notice, and information provision — than on the technical evaluation of explanations. The NIST AI RMF explicitly addresses model explanation through MEASURE 2.9, whereas none of the five instruments prescribes a specific XAI method or a universal numerical explanation-quality threshold; methods such as LIME, SHAP, Integrated Gradients, Grad-CAM, and TCAV therefore enter the mapping as literature-supported technical candidates.

Instrument	Key clause(s)	What must be produced	Candidate methods	Sources
EU AI Act (Reg. 2024/1689)	Art. 13(1–3); Art. 14; Art. 50 — transparency obligations for certain AI systems and AI-generated or manipulated content	Instructions for use; capabilities, limitations and intended purpose disclosed; information supporting deployer understanding and human oversight; AI-content disclosure	LIME [11]; SHAP [12]; Integrated Gradients [56]; Grad-CAM [57]; TCAV [26]; technical documentation addressing the minimum information required by Annex IV	[11, 12, 26, 28, 56, 57]
NYC Local Law 144 of 2021	§20-871(b)(1–2); Rules §5-303	Candidate notice ≥10 business days before AEDT use with specified content; public posting of bias audit summary	Documentation and disclosure compliance (binary checks); no model-side XAI required by the law	NYC Local Law 144 and DCWP Final Rule [39]
NIST AI RMF 1.0	Measure 2.8 (transparency risks); Measure 2.9 (model explained and documented)	Transparency risks examined and documented; model explanations produced and reviewed	Post-hoc XAI (LIME [11], SHAP [12], TCAV [26]); Quantus evaluation framework [28]	[11, 12, 26, 28, 30, 31]
OECD AI Principles	Principle 1.3 — transparency and explainability	Principle-level commitment to transparency and explainability; no operative method or disclosure procedure specified	None specified; values-level orientation only	OECD AI Principles (2019, rev. 2024)
ISO/IEC 23894:2023	Transparency risk-treatment clauses (mapped from ISO 31000)	Transparency risks identified, treated, and monitored; controls documented	IEEE 7001-2021 (transparency levels per stakeholder); post-hoc XAI for model-side controls [26, 28]	IEEE 7001-2021; [26, 28]

Table 4 — Transparency: operative clauses and candidate methods.

5.3 Accountability coverage and operational mapping across the shortlist

Accountability is realised primarily through documentation artefacts, audit mechanisms and clear allocation of responsibilities. The AI Act distributes obligations across the value chain (providers, deployers, importers and distributors) and requires post-market monitoring, risk-management systems and documented conformity assessment. Annex IV specifies the minimum information to be included in the technical documentation, while Article 17 requires providers of high-risk AI systems to establish a quality management system. ISO/IEC 42001 provides a complementary AI management-system standard that can support such governance and documentation practices [38].

In the NIST AI RMF, accountability is embedded in the Governance function and covers policies, roles and audit procedures. The Generative AI Profile [37] provides suggested actions for documenting and managing generative-AI risks. The Critical Infrastructure concept note [47] announces the development of a sector-specific AI RMF Profile; it is not itself a completed profile or an operational requirements document. New York’s law mandates independent audits and public disclosure of results and provides for civil penalties [39]; the 2025 State Comptroller audit identified weaknesses in enforcement and recommended stronger complaint handling, technical review and proactive identification of potential non-compliance [48].

The OECD principles continue to call for accountability but leave implementation to national regulators and technical frameworks.

Accountability evaluation is primarily qualitative and process-oriented. It considers not only whether relevant artefacts and governance mechanisms exist, but also whether they are sufficiently complete, current, traceable, and suitable for the intended purpose. Relevant evidence includes model cards, datasheets, audit logs, risk registers, responsibility assignments, audit procedures, and monitoring records. The presence of an artefact is therefore treated as a necessary but not sufficient indicator of accountability; where feasible, its completeness and evidentiary quality should also be assessed.

A recent U.S. procurement directive, OMB M-26-04 (December 2025), requires federal agencies procuring LLMs to request a minimum transparency package from vendors, including acceptable-use policies, model/system/data cards, end-user resources, and feedback mechanisms. Additional model-evaluation evidence may be requested depending on the intended use. In D2.1, this source is used only as a supplementary example of procurement-driven accountability evidence, not as a binding or core requirement for EXCALIBUR.

Table 5 consolidates the accountability provisions and evidence requirements associated with the five shortlisted instruments. Unlike Fairness and XAI, Accountability is represented primarily through documentation, audit procedures, responsibility allocation, risk-management records, and monitoring evidence rather than through numerical metrics. The AI Act provides the strongest source-defined documentation and governance requirements, NYC Local Law 144 contributes an independent-audit and public-disclosure model, while the NIST AI RMF, ISO/IEC 23894, and OECD AI Principles provide progressively more process-oriented or principle-level accountability guidance.

Instrument	Key clause(s)	What must be produced	Candidate methods / artefacts	Sources
EU AI Act (Reg. 2024/1689)	Art. 11; Art. 12; Art. 17; Arts. 16-27; Art. 43; Art. 47; Annex V; Art. 72	Technical documentation (Annex IV); automated event logs; QMS; conformity assessment; EU Declaration of Conformity (Annex V); post-market monitoring system	Model cards [15]; datasheets for datasets [16]; SMOACTR internal audit pipeline [32]; Annex IV technical-documentation requirements; event-log schema	Annex IV; [15, 16, 32]
NYC Local Law 144 of 2021	§20-871; Rules §5-302, §5-303	Annual independent bias audit; audit summary publicly posted; audit documentation maintained	Third-party audit procedure; DCWP-aligned audit report format; website posting	NYC Local Law 144 and DCWP Final Rule [39]
NIST AI RMF 1.0	Govern function; Govern 2.1 (roles and communication); Manage 4 (incident response, post-deployment monitoring)	Governance documentation; roles and responsibilities assigned; incident and error logs maintained	NIST AI RMF Playbook suggested actions and documentation practices	NIST AI RMF Playbook (2023)
OECD AI Principles	Principle 1.5 — accountability	Principle-level commitment to accountability; no operative artefact or procedure specified	None specified; values-level orientation only	OECD AI Principles (2019, rev. 2024)
ISO/IEC 23894:2023	AI risk management clauses (AI-scoped from ISO 31000)	AI risk management records, risk owners, review cadence, and residual risk documentation; can be complemented by ISO/IEC 42001 management-system controls.	ISO/IEC 23894 risk register; ISO/IEC 42001 audit trail (as supporting source)	ISO/IEC 23894; ISO/IEC 42001 as supporting source

Table 5 — Accountability: operative clauses and candidate artefacts.

The accountability frameworks identified in the scientific literature are operationalised separately in Table 8 and documented in Appendix B. They support the technical implementation of the five-instrument mapping but do not constitute additional core regulatory or standardisation instruments.

5.4 Coverage and silences

These developments do not change the basic finding that many regulatory and standardisation instruments remain under-specified with respect to technical metrics. The 2025 GPAI Code of Practice and training-content summary template [36], together with the 2026 Article 50 Guidelines and Transparency Code [54, 55], provide additional implementation guidance on transparency and documentation but do not define fairness metrics or universal fairness thresholds. The NIST Generative AI Profile highlights generative-AI risks, while the Critical Infrastructure concept note describes a forthcoming profile; neither document prescribes specific fairness metrics or thresholds [37, 47]. ISO/IEC 42001 standardises governance requirements but leaves metric selection to organisations. The mapping tables in §5.1–§5.3 and §6 therefore continue to rely on the literature to suggest appropriate metrics and thresholds. Where instruments provide no direct guidance, we record that absence explicitly and distinguish metric/control provenance from threshold status in accordance with the conventions of §6.1.

6 Metrics, Thresholds and Evidence-Based Justification

This section operationalises the mapping developed in Section 5 by connecting the five shortlisted instruments — the EU AI Act, NYC Local Law 144, the NIST AI Risk Management Framework, the OECD AI Principles, and ISO/IEC 23894:2023 — to concrete metrics, controls, evidence artefacts and threshold sources. Its purpose is not to propose a new set of arbitrary thresholds for trustworthy AI. Instead, it makes explicit which thresholds are directly grounded in a regulatory or standardisation source, which are supported by the scientific literature, which are project-defined for EXCALIBUR experimentation, and where no established threshold exists. The analysis also takes into account post-2024 developments, such as emerging fairness metrics for generative AI models, enterprise-level explainability requirements and U.S. procurement directives for large language models (LLMs).

The shortlisted instruments do not operate at the same level of specificity. Some instruments, such as the AI Act, impose binding obligations on data governance, documentation, logging, transparency and post-market monitoring. However, they generally do not specify the exact fairness metric, explainability metric, or numerical cut-off that should be used. Recent implementation guidance for the AI Act, including the transparency provisions applicable from August 2026, the voluntary GPAI Code of Practice, and the public training-content summary template required under Article 53(1)(d), emphasises documentation and training-data summaries rather than prescribing metrics [9, 36]. Other instruments, such as the NIST AI RMF and ISO/IEC 23894, provide structured risk-management and governance guidance, but they remain non-binding and mostly process-oriented. The NIST Generative AI Profile (July 2024) [37] provides suggested actions for managing and documenting generative-AI risks without prescribing specific fairness or explainability thresholds. The Critical Infrastructure concept note (April 2026) [47] announces a forthcoming sector-specific profile and is therefore used only as contextual evidence of ongoing NIST guidance development. NYC Local Law 144 is the most operational fairness-audit source in the shortlist because the DCWP Final Rule defines the required calculations and provides worked numerical examples for selection rates, scoring rates, impact ratios, and intersectional reporting [39]. The OECD AI Principles provide a principle-level international reference point, but they do not define technical metrics or thresholds. A U.S. memorandum (OMB M-26-04, December 2025) requires federal agencies procuring LLMs to request a minimum transparency package and allows additional evaluation and control evidence to be requested where justified by the intended use. Agencies were required to update their procurement policies and procedures by 11 March 2026 [40].

For this reason, the tables in this section should be read as traceability tables. Each row records (i) the external instrument or scientific publication that creates, motivates, or supports the relevant requirement, (ii) the metric or control used for operationalisation and its provenance, (iii) the available threshold or interpretation criterion and its status or supporting source, and (iv) the main limitation. Where an instrument does not define a technical metric or control, the corresponding operational element is identified as a literature-supported candidate rather than as an instrument-defined requirement. Where no externally justified threshold exists, the table records this explicitly instead of introducing an unsupported cut-off. Where the regulation or standard does not define a metric, the metric is marked as a candidate implementation derived from the literature corpus and the accompanying metric catalogue. Where no threshold exists, the row does not invent one; it records the absence of an established threshold and explains the limitation.

6.1 Threshold typology and table conventions

The threshold typology introduced in Section 2.4 is applied throughout this section. Four categories are used. Source-defined thresholds are thresholds, criteria or mandatory checks explicitly defined by one of the shortlisted instruments or by a directly referenced regulatory or standardisation source. Literature-supported thresholds are values, interpretation rules or evaluation protocols proposed or used in the scientific literature. Project-defined operational thresholds are practical interpretation rules that EXCALIBUR may propose for in-lab or in-the-wild experimentation, subject to consortium approval, pre-registration, and empirical validation; they are not regulatory or literature-derived thresholds. None established means that the metric or control can be reported, but no universal or externally justified cut-off exists. Given the rapid emergence of new fairness metrics for LLMs and domain-specific evaluation frameworks, it is increasingly important to treat thresholds as dynamic and context-dependent rather than fixed rules. Separately from threshold status, the Metric/control provenance column records whether the operational element itself is instrument-defined, literature-supported, or project-defined. This distinction prevents the provenance of a metric or control from being conflated with the separate question of whether an externally justified acceptance threshold or interpretation criterion exists.

A further distinction concerns coverage. Tables 6–8 operationalise a selected subset of metrics, methods, and artefacts — those carried forward for EXCALIBUR with explicit threshold sources — and Appendix A gives the canonical formulation of the group-fairness criteria among them. The complete set of fairness metrics, explainability evaluation methods, and accountability instruments identified in the Task 2.1 literature review is catalogued in Appendix B, which records for each entry its defining reference and whether it was adopted into a table or retained for reference only. The main text therefore reflects a deliberate selection from the surveyed landscape rather than its full extent.

Two conventions apply to the tables. First, when a row links a metric to a regulation or standard, this does not necessarily mean that the source explicitly names that metric. For example, the AI Act requires training data to be examined in view of possible biases, but it does not name demographic parity, equalised odds or disparate impact. These metrics are therefore treated as possible technical implementations of a broader obligation. New fairness metrics such as the Uncertainty-Aware Fairness (UCerF) metric proposed for LLMs incorporate model confidence and provide a finer-grained evaluation of internal bias [41]. When these new metrics are used, the traceability table records their source, purpose and any assumptions about uncertainty or calibration. Second, when a threshold is imported from a document outside the five shortlisted instruments, the threshold source is explicitly named and the limitation column flags the indirection. Among the five shortlisted instruments, NYC Local Law 144, together with the DCWP Final Rule, provides the most explicit source-defined fairness-audit calculation procedure and worked examples. These examples are used as source-defined operational illustrations. For the remaining four instruments, candidate metrics and controls are supplied by the scientific literature and are treated as literature-supported mappings rather than as metrics explicitly prescribed by the source.

The tables use the following columns: Source / clause, Metric or control, Metric/control provenance, Threshold / criterion, Threshold status / source, and Limitation. This separates the provenance of an operational metric or control from the distinct question of whether an externally justified acceptance threshold or interpretation criterion exists. This structure is retained across fairness, transparency / XAI, and accountability so that all candidate metrics and controls can be compared using the same traceability logic.

A colour convention also applies from this section onward. Tables 6–8 and the taxonomy in Section 7 use pillar-specific shading as a secondary visual cue for Fairness, Transparency/XAI, and Accountability. Colour is never the sole indicator of category.

Fairness	Transparency / XAI	Accountability
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6.2 Fairness metrics: thresholds and evidence-based limitations

Fairness is the most metric-rich pillar, but it is also the pillar where threshold interpretation is most fragile. The scientific literature provides many fairness definitions: independence-family metrics such as Statistical Parity and Disparate Impact; separation-family metrics such as Equalised Odds and Equal Opportunity; sufficiency-family metrics such as Predictive Parity and Calibration within Groups; individual fairness metrics; subgroup fairness metrics; and causal or counterfactual fairness definitions. Recent work has extended these families to generative AI and large language models. The UCerF metric introduces uncertainty-aware fairness by weighting misclassification rates with model confidence, thereby capturing bias that standard metrics may overlook [41]. In the healthcare domain, a five-step framework has been proposed to audit bias in clinical LLMs, emphasising stakeholder engagement, model calibration, scenario-based testing and evaluation of model outputs rather than hyperparameters [42]. A comprehensive survey of LLM evaluation [43] organises assessment across capability, robustness, bias and safety dimensions and across multiple benchmarks, combining automatic and human evaluation.

The AI Act creates a strong fairness-related obligation through its requirements on data governance and bias examination, especially for high-risk systems. NIST AI RMF and ISO/IEC 23894 support fairness through measurement, risk identification and risk treatment. The OECD AI Principles support fairness at the level of values and human-centred AI. NYC LL 144 is the most concrete fairness-audit source in the shortlist because it requires bias audits and reporting for automated employment decision tools. Among the five shortlisted instruments, NYC Local Law 144 provides the most concrete operational fairness-audit criteria through source-defined calculations and worked examples of selection rates, scoring rates, and impact ratios [39]. The remaining instruments establish fairness-related obligations or risk-management expectations but rely on scientific literature for their operationalisation. Procurement-focused sources such as OMB M-26-04 do not prescribe fairness thresholds and are therefore not used as threshold-setting sources in this section.

Table 6 consolidates the fairness metrics and controls carried forward for EXCALIBUR and records, for each entry, its provenance, available threshold or interpretation criterion, threshold status, and principal limitation. Three observations are particularly important. First, source-defined numerical fairness material remains scarce. Within the five-instrument shortlist, the DCWP Final Rule is the only instrument that provides worked calculations of selection rates, scoring rates, and impact ratios [39]. These examples operationalise the audit and reporting procedure; they do not establish a universal fairness cut-off. Second, many fairness metrics can be calculated, but most should be interpreted comparatively and contextually rather than as pass/fail thresholds. Third, EXCALIBUR should avoid presenting fairness as a single score. Instead, the platform should report the metric family, protected attribute, affected group or subgroup, decision task, instrument or literature basis, worldview commitment, known

incompatibility constraints, and any uncertainty or calibration considerations. For LLM-based systems, additional bias and safety evaluation may be reported using the relevant literature, without treating emerging metrics as source-defined requirements [41–43].

Source / clause	Metric or control	Metric/control provenance	Threshold / criterion	Threshold status / source	Limitation
EU AI Act Art. 10 — data governance and bias examination	Statistical Parity / Demographic Parity; Disparate Impact Ratio; Equalised Odds; Equal Opportunity; Predictive Parity; Calibration within Groups; subgroup fairness metrics	Instrument-defined bias-examination obligation under the AI Act [9]; candidate fairness metrics are literature-supported [14, 21, 23–25].	No source-defined numerical threshold. The metric family must be selected and justified according to the decision task, protected attribute, affected population, harm model, and fairness worldview.	No established universal threshold. Any subgroup γ tolerance must be explicitly identified as literature-supported or project-defined.	The AI Act requires bias examination but does not name a fairness metric or cut-off. Metric choice encodes a fairness worldview and may conflict with other metric families.
NYC Local Law 144 — bias audit for AEDTs	Selection rate; scoring rate; impact ratio; sex, race/ethnicity, and intersectional reporting	Instrument-defined audit and calculation requirements under NYC Local Law 144 and the DCWP Final Rule [39].	Independent bias audit conducted within the preceding year; selection rates or scoring rates, as applicable, and corresponding impact ratios calculated and reported for required categories; audit summary publicly disclosed.	Source-defined procedural and calculation criterion — NYC Local Law 144 / DCWP Final Rule [39]. No universal fairness pass/fail cut-off is established.	The worked examples define calculation and reporting procedures. Interpretation of observed disparities depends on employment context, dataset composition, sample size, and the wider legal and organisational assessment.
NIST AI RMF — fairness measurement and risk management functions	Demographic parity difference; Disparate Impact; Equalised Odds; Equal Opportunity; subgroup fairness diagnostics; calibration reporting	NIST provides the measurement and risk-management guidance; the candidate fairness metrics are literature-supported [14, 21, 23–25].	No NIST-defined numerical threshold. Metrics should be reported with context, protected attribute, decision task, and selected fairness worldview.	None established by NIST; interpretation is literature-supported and context-dependent.	NIST provides measurement and governance structure but does not prescribe fairness metrics or numerical cut-offs.
OECD AI Principles — inclusive growth, human-centred values, fairness	High-level fairness assessment; candidate use of group, individual, and subgroup fairness metrics	Principle-level instrument; candidate technical metrics are literature-supported rather than prescribed by OECD.	No threshold.	None established — OECD AI Principles [50].	OECD supports fairness as a value but does not provide operative metrics or thresholds.
ISO/IEC 23894:2023 — AI risk management	Bias as AI-related risk; candidate use of fairness metrics as risk indicators and treatment evidence	Standard-based risk-management control; candidate technical metrics are literature-supported [14, 24].	No ISO-defined numerical threshold. DI or subgroup criteria may be used only if justified externally.	None established by ISO/IEC 23894; any imported criterion must be explicitly attributed to its external source.	ISO/IEC 23894 supports risk identification, evaluation, treatment, and monitoring but does not prescribe fairness metrics or thresholds.
Cross-cutting fairness interpretation	Incompatibility of Independence, Separation, and Sufficiency families	Literature-supported conceptual limitation [17, 18, 21, 23, 25].	No universal threshold; metric families cannot generally all attain their ideal values simultaneously when base rates differ.	Literature-supported limitation; no acceptance threshold exists.	Any fairness report must state the selected metric family and worldview. A single fairness score would conceal incompatibilities.
Subgroup fairness / fairness gerrymandering	γ -Statistical Parity Subgroup Fairness; γ -False Positive Subgroup Fairness; intersectional impact ratios	Literature-supported metrics and experimental protocol [14].	The cited study explores γ values from 0 to 0.029 in a specific experimental setting. This is not a general acceptance range.	No established threshold. The experimental γ sweep is literature-supported [14]; any EXCALIBUR alert/tolerance would be project-defined and require validation.	Subgroup auditing is computationally demanding and depends on the predefined subgroup class G. Ex-post subgroup selection risks overfitting.
Causal and counterfactual fairness	Counterfactual Fairness; Path-specific Counterfactual Fairness; Direct Counterfactual Fairness; Counterfactual Equalised Odds; Principal Fairness	Literature-supported fairness definitions [13, 21, 23, 25].	No general threshold.	None established.	Requires a causal graph, structural causal model, or potential-outcomes assumptions. Not assessable from observational data alone.

Table 6 — Fairness metrics and controls: provenance, threshold status, and limitations.

The metrics in Table 6 are defined operationally in Appendix A, which gives the canonical formulation of each criterion—statistical parity, equalised odds, equal opportunity, predictive-value parity, calibration, and class balance—and, where available, identifies a corresponding implementation or diagnostic in the AIF360 toolkit [46]. This allows the obligations mapped here to be measured rather than only asserted. The mutual incompatibility of these criteria under realistic base rates, and the resulting need to select metrics per use case rather than report them all, follows the comparative analysis of Garg, Villasenor & Foggo [45].

6.3 Transparency and XAI metrics: thresholds and limitations

Transparency in the shortlisted instruments has two dimensions. The first is process-side transparency, which concerns disclosure, documentation, information provision, notice and traceability. The second is model-side transparency, which concerns explainability and XAI: whether a model's behaviour or output can be explained in a way that is faithful, useful and understandable to the relevant stakeholder. Beyond local feature-attribution scores such as SHAP or LIME, training-data attribution methods based on influence functions [44] trace a prediction back to the training examples most responsible for it, complementing feature-level explanations with data-level evidence; audit-trail completeness and contestability mechanisms are treated as accountability evidence in §6.4. As a supplementary U.S. procurement example, OMB M-26-04 requires federal agencies procuring LLMs to request a minimum transparency package, including acceptable-use policies, model/system/data cards, end-user resources, and feedback mechanisms; additional evaluation evidence may be requested depending on the intended use [40], while the EU AI Act's transparency obligations emphasise summarising training data and marking AI-generated content [9, 36].

The shortlisted instruments are stronger on process-side transparency than on XAI evaluation. The AI Act requires technical documentation, record-keeping and information to deployers, but it does not require LIME, SHAP, TCAV, counterfactual explanations or any specific explanation-quality metric. The NIST AI RMF is voluntary and calls for AI models to be explained, validated and documented, while leaving the choice of methods open. ISO/IEC 23894 frames transparency as part of AI risk management. NYC LL 144 requires notice and publication of bias-audit information but does not require model explanations. OECD AI Principles support transparency and explainability at the principle level. The NIST Generative AI Profile [37] and the LLM evaluation literature [43] stress that evaluation should encompass accuracy, coherence, factuality, safety and bias. Audit trails and contestability are treated in this deliverable as accountability evidence rather than as model-side XAI outputs, and are addressed under accountability in §6.4.

For this reason, the transparency/XAI table relies heavily on the scientific literature. The most useful candidate evidence comes from concept-based explanations such as TCAV; explanation-faithfulness metrics such as Infidelity and ROAR; robustness metrics such as Max-Sensitivity; randomisation sanity checks; human-centred instruments such as the Explanation Satisfaction Scale; and recently proposed evaluation protocols for generative models that combine automatic metrics with human evaluation. A comprehensive survey of LLM evaluation [43] stresses that assessment should span factuality, safety and bias and should combine benchmark-based and human evaluation rather than rely on any single automatic score.

Table 7 consolidates the Transparency/XAI metrics and controls selected for operationalisation and records their provenance, threshold or interpretation criteria, threshold status, and limitations. Three observations are particularly important. First, no shortlisted instrument defines a specific XAI method or a universal numerical threshold for explanation quality. EXCALIBUR should not claim that a SHAP value, a LIME explanation, or a TCAV score is compliant by itself. Compliance or alignment can only be argued through documented evaluation evidence. Second, XAI evaluation is moving towards multi-dimensional assessment: lower Infidelity and Max-Sensitivity, successful randomisation checks, performance above random baselines in ROAR, and statistically significant TCAV results [26–31]. Where feasible, influence functions may provide supplementary training-data attribution evidence [44]. Audit-trail completeness and contestability are treated separately as accountability controls in §6.4. Third,

human-centred explanation measures are necessary for EXCALIBUR because the project targets non-expert stakeholders, but they must be separated from faithfulness metrics. A user may be satisfied with an explanation that is incomplete or over-simplified; therefore, EXCALIBUR should report both technical explanation quality and user-side evaluation. When working with generative models, user studies should also examine whether explanations help users detect hallucinations or unreliable outputs and whether they understand the provenance of training data and model biases [43].

Source / clause	Metric or control	Metric/control provenance	Threshold / criterion	Threshold status / source	Limitation
EU AI Act Arts. 11–13 — technical documentation, record-keeping, transparency and information provision	Candidate XAI methods: LIME [11], SHAP [12], Integrated Gradients [56], Grad-CAM [57], TCAV [26], counterfactual explanations; XAI evaluation reports	Instrument-defined transparency/documentation requirements under the AI Act [9]; specific XAI methods and evaluation procedures are literature-supported [11, 12, 19, 26–31, 56, 57].	No AI Act-defined XAI threshold. XAI methods must be documented and evaluated rather than assumed faithful.	No established source-defined XAI threshold. Method/evaluation criteria are literature-supported.	The AI Act requires transparency but does not prescribe XAI methods or numerical explanation-quality thresholds.
EU AI Act Art. 14 — human oversight	Explanation Satisfaction Scale; Explanation Goodness Checklist; mental-model / trust / usability measures	Instrument-defined human-oversight requirement; candidate human-centred evaluation instruments are literature-supported [29].	ESS is a validated instrument; no universal pass cut-off. Comparative interpretation across explanation variants is recommended.	No established universal threshold; literature-supported evaluation protocol [29].	Human-centred scores are task- and stakeholder-dependent. High satisfaction does not establish explanation faithfulness.
NYC Local Law 144 — candidate notice and public audit information	Candidate notice period; public availability of bias-audit report; disclosure of AEDT use and evaluated characteristics	Instrument-defined procedural transparency controls [39].	Notice at least 10 business days before AEDT use; annual independent bias audit.	Source-defined criterion — NYC Local Law 144 / DCWP Final Rule [39].	Strong on procedural notice and public audit disclosure; weak on technical model explainability.
NIST AI RMF — explainability, interpretability, documentation, monitoring	XAI evaluation suite: faithfulness, robustness, randomisation, complexity, localisation, axiomatic checks	NIST provides the explainability and documentation guidance; specific evaluation metrics and protocols are literature-supported [27, 28, 30, 31].	No NIST-defined threshold. Metric-specific directions apply: Infidelity ↓; sensitivity/local-Lipschitz ↓; faithfulness correlation ↑; randomisation checks use a pre-specified comparison protocol.	No NIST numerical threshold; literature-supported relative criteria [27, 28, 30, 31].	Direction indicators are not universal acceptance thresholds; individual metrics may have different directions.
OECD AI Principles — transparency and explainability	High-level explainability and disclosure requirement; possible human-centred explanation evaluation	Principle-level instrument; candidate evaluation instruments are literature-supported [29].	No threshold.	None established — OECD AI Principles [50].	Provides a values baseline but no operative XAI metrics.
ISO/IEC 23894:2023 — transparency as AI risk treatment	Transparency risk controls; stakeholder-specific transparency levels if supported by additional standards	Standard-based risk-management control; additional operationalisation may rely on supporting standards such as IEEE 7001.	No ISO/IEC 23894-defined numerical threshold.	None established by ISO/IEC 23894; qualitative criteria may be imported from explicitly identified supporting standards.	ISO/IEC 23894 does not operationalise XAI quality by itself.
TCAV — concept-based explanation	TCAV score; Concept Activation Vector quality	Literature-supported method [26].	Statistical significance testing against no-concept influence; protocol depends on TCAV setup.	Literature-supported statistical evaluation protocol [26]; no universal numerical acceptance threshold.	Requires user-defined concepts and positive/negative concept examples. Concept quality affects validity.
Infidelity and Max-Sensitivity	Infidelity for faithfulness; Max-Sensitivity for robustness	Literature-supported metrics [27].	Lower is better; use relative comparison between methods or explanation variants.	Literature-supported relative criterion [27]; no absolute cut-off.	Sensitivity alone can favour trivial constant explanations and should be interpreted jointly with faithfulness.
Model Parameter Randomisation Test and Data Randomisation Test	Sanity checks for attribution methods	Literature-supported diagnostic protocol [30].	Use a pre-specified comparison protocol to verify that attribution maps respond materially to parameter or label randomisation.	No universal similarity threshold [30]. Any explicit acceptance rule adopted by EXCALIBUR would be project-defined.	Results depend on similarity metric, randomisation design, and the chosen project acceptance rule.
ROAR faithfulness benchmark	Remove And Retrain evaluation	Literature-supported benchmark [31].	Explanation method should outperform a random-feature-ranking baseline, particularly at substantial removal fractions.	Literature-supported relative benchmark [31]; no universal numerical cut-off.	Computationally expensive; some saliency methods may perform no better than random baselines.
Explanation Satisfaction Scale	Human-centred satisfaction with explanations	Literature-supported evaluation instrument [29].	Use comparative interpretation rather than absolute pass/fail.	No established universal threshold [29].	Measures perceived understanding and usefulness, not necessarily causal/model faithfulness.

Table 7 — Transparency/XAI metrics and controls: provenance, threshold status, and limitations.

6.4 Accountability controls: artefacts, auditability, and evidence quality

Accountability differs from fairness and XAI because it is usually not evaluated through continuous mathematical metrics. It is mostly assessed through evidence artefacts, logs, documentation, governance procedures, audit reports, responsibility allocation and monitoring processes. Accordingly, the accountability table uses qualitative, process-completeness, and evidence-quality criteria. Binary presence checks may be retained where an instrument explicitly requires the existence of an artefact or process, but presence alone is not treated as sufficient evidence of effective accountability. Recent policies and standards have expanded the list of required artefacts. The OMB M-26-04 memo requires U.S. federal agencies procuring LLMs to request a minimum transparency package from vendors, including acceptable-use policies, model/system/data cards, end-user resources, and feedback mechanisms. It also allows agencies to request additional evidence, such as model evaluations or enterprise-level controls, where this is justified by the intended use [40]. ISO/IEC 42001:2023 introduces a formal Artificial Intelligence Management System (AIMS) with requirements for risk management, lifecycle management and third-party oversight [38]. Complete audit trails and contestability mechanisms are likewise treated as core accountability evidence, consistent with the SMACTR internal-audit framework [32] and the AIMS oversight requirements of ISO/IEC 42001 [38]. These additions mean that accountability evaluation must cover the presence and quality of documentation artefacts as well as evidence that auditability and contestability mechanisms are in place.

The AI Act remains the strongest source-defined accountability instrument in the shortlist. It requires technical documentation, record-keeping, risk management, post-market monitoring and conformity assessment. New EU implementation guidance for general-purpose AI models, published in July 2025, includes a voluntary Code of Practice and a template for public training-data summaries [36], supporting the implementation of the transparency, copyright, and documentation obligations applicable to general-purpose AI model providers. NIST AI RMF emphasises governance structures and auditability but leaves metrics to implementers; ISO/IEC 23894 addresses risk management and process controls; ISO/IEC 42001 codifies a management-system approach; NYC LL 144 mandates independent bias audits and disclosure; the OECD Principles urge accountability but remain high-level. The OMB M-26-04 memorandum makes specified transparency documentation and, where justified by the intended use, additional model-evaluation evidence part of federal LLM procurement, illustrating how accountability requirements can extend into supply-chain relationships [40].

Given the above, accountability controls in EXCALIBUR should include: presence of model cards and datasheets; audit logs; risk registers; compliance with Code of Practice or procurement requirements; records of evaluation artefacts and feedback mechanisms; evidence of responsibility allocation; and documentation of post-deployment monitoring. For EXCALIBUR, the useful lesson is not alignment with the U.S.-specific Unbiased AI Principles, but the broader accountability pattern: LLM-generated explanations and fairness reports should be anchored in verifiable documentation, evaluation evidence, provenance information, and feedback mechanisms.

Table 8 consolidates these accountability controls and distinguishes source-defined requirements from literature-supported and project-defined evidence artefacts. The table highlights that accountability is primarily assessed through the presence, completeness, quality, and traceability of documentation and governance processes rather than through universal numerical thresholds. The strongest source-defined controls originate from the AI Act and NYC Local Law 144, while model cards, datasheets, internal-audit frameworks, responsible-AI process metrics, and LLM-specific audit structures provide complementary literature-supported evidence for EXCALIBUR.

Source / clause	Metric or control	Metric/control provenance	Threshold / criterion	Threshold status / source	Limitation
EU AI Act Art. 11 and Annex IV	Technical documentation	Instrument-defined accountability/documentation control [9].	Mandatory presence; content must satisfy applicable Annex IV information requirements.	Source-defined qualitative criterion — Art. 11 and Annex IV.	Annex IV specifies minimum information but not a fixed machine-readable XAI/fairness evidence package.
EU AI Act Art. 12	Automatic event logs	Instrument-defined logging requirement [9].	High-risk AI systems must technically allow automatic recording of relevant events over their lifetime.	Source-defined qualitative/process criterion — Art. 12.	Log fields are not fully standardised; EXCALIBUR must determine which fairness/XAI events need recording.
EU AI Act Art. 17	Quality Management System	Instrument-defined organisational control [9].	Mandatory presence for providers of high-risk AI systems.	Source-defined criterion — Art. 17.	Does not prescribe specific fairness or XAI metrics.
EU AI Act Art. 43; Art. 47; Annex V	Conformity assessment and EU Declaration of Conformity	Source-defined conformity controls [9].	Applicable conformity-assessment route and EU Declaration of Conformity required where applicable.	Source-defined criterion — Arts. 43, 47 and Annex V.	Does not define a fixed fairness/XAI evidence package.
EU AI Act Art. 72	Post-market monitoring	Instrument-defined monitoring requirement [9].	Presence of the required post-market monitoring system.	Source-defined process criterion — Art. 72.	Does not prescribe which technical fairness/XAI metrics must be monitored.
EU AI Act Arts. 16–27	Responsibility allocation across provider, deployer, importer, distributor, and other actors	Instrument-defined role allocation [9].	Obligations assigned according to actor role.	Source-defined role-based criterion — Arts. 16–27.	Correct role identification may be complex in open-source, third-party-model, or LLM pipelines.
NYC Local Law 144	Annual independent bias audit and public disclosure	Instrument-defined audit/disclosure control [39].	Independent bias audit at least annually; results publicly available; candidate notice before AEDT use.	Source-defined procedural criterion [39].	Narrow jurisdiction/domain and limited specification of overall fairness success criteria.
NIST AI RMF — Govern, Map, Measure, Manage	Governance documentation; risk register; accountability roles; impact assessment; monitoring evidence	Guidance-based controls under the NIST AI RMF [49].	No fixed numerical threshold; evidence should demonstrate implementation of relevant RMF functions.	No universal threshold; organisation-specific qualitative/process criteria.	Voluntary framework; frequency and completeness criteria are organisation-defined.
OECD AI Principles — accountability	Accountability mechanisms, contestability, redress, responsible actors	Principle-level instrument [50].	No threshold.	None established.	High-level principle without an implementation schema.
ISO/IEC 23894:2023	AI risk-management process; identification, analysis, evaluation, treatment, and monitoring records	Standard-based risk-management control [51].	Presence and substantive implementation of documented risk-management process and treatment evidence.	Standard-based qualitative/process criterion; no numerical threshold.	Does not define specific fairness/XAI metrics or monitoring thresholds.
Model Cards	Model-level documentation: intended use, performance, limitations, ethical considerations, fairness evidence	Literature-supported documentation artefact [15].	Presence and completeness of model card.	No universal quality threshold [15].	Presence alone does not establish documentation quality or compliance.
Datasheets for Datasets	Dataset-level documentation: motivation, composition, collection, preprocessing, uses, maintenance	Literature-supported documentation artefact [16].	Presence and completeness of datasheet.	No universal quality/completeness threshold [16].	Useful for governance evidence but no universal scoring or certification rule exists.
SMACTR internal audit framework [32]	Five-phase internal audit pipeline and associated artefacts	Literature-supported audit framework [32].	Evidence of the five main phases and maintenance of traceable audit artefacts across the lifecycle.	Literature-supported process criterion [32]; no numerical acceptance threshold.	Internal auditing does not replace independent external audit; adaptation to federated/LLM pipelines remains open.
Responsible AI Metrics Catalogue [33]	Process metrics across Responsibility, Auditability, and Redressability	Literature-supported process catalogue [33].	Applicable process metrics should be substantively completed and evidenced.	Literature-supported process criterion; no validated universal numerical threshold [33].	Self-assessment instrument; operational verification mechanisms are not fully specified.
Three-layered LLM audit blueprint [34]	Governance audit + model audit + application audit	Literature-supported LLM audit framework [34].	Coverage of all three layers; upper-layer outputs inform lower layers; application-level evaluation combines ex-ante and continuous ex-post assessment.	Literature-supported process criterion [34]; no universal numerical threshold.	Auditor arrangements/licensing are unsettled; continuous application-level monitoring is required for dynamic LLM use.
XAI / fairness audit report	Consolidated evidence package: metric values, interpretation criteria, provenance, model/dataset versions, protected attributes, limitations	Mixed provenance: instrument-defined where an external instrument requires an audit/evidence artefact; otherwise project-defined for EXCALIBUR.	Mandatory only where linked to a source-defined audit obligation; otherwise an EXCALIBUR evidence artefact.	Source-defined where explicitly required; otherwise project-defined.	Candidate literature metrics must not be represented as legally mandated thresholds.

Table 8 — Accountability controls: provenance, evidence criteria, threshold status, and limitations.

6.5 Candidate project-defined interpretation rules for later EXCALIBUR experimentation

This subsection does not establish thresholds that are already adopted by EXCALIBUR. It identifies candidate interpretation rules that may be specified in later experimental or validation activities where no external threshold exists. Any such rule must be task-specific, documented in advance, empirically validated, versioned, and clearly labelled as project-defined rather than regulatory or literature-derived.

Project-defined operational thresholds in D2.1 are candidate rules for later EXCALIBUR experimentation; none is adopted by this deliverable. Possible candidates include: (a) a pre-registered subgroup-fairness tolerance selected per task and dataset; (b) pre-specified model/data-randomisation and ROAR diagnostic protocols; (c) comparative interpretation of Explanation Satisfaction Scale scores; (d) reporting of calibration and uncertainty measures where relevant; and (e) avoidance of a universal group-fairness cut-off. UCerF or similar uncertainty-aware metrics [41] and training-data-attribution methods [44] may be evaluated where relevant and feasible, but their inclusion, acceptance criteria, and alert levels require consortium approval, pre-registration, and empirical validation in the relevant downstream work package.

These candidates are not regulatory or literature-derived thresholds and do not constitute EXCALIBUR-wide policy. Any rule later adopted should record its purpose, metric, direction, tolerance, task/dataset, evidence source, validation results, version, and approving body.

6.6 Synthesis: what is grounded, what is supported, and what remains open

Taken together, Tables 6–8 show that source-defined numerical thresholds remain rare across the five shortlisted instruments and that the three pillars require different forms of operational evidence. Fairness criteria are context-dependent; even NYC Local Law 144, the most operational fairness-audit source in the shortlist, defines calculations and reporting procedures rather than a universal fairness cut-off [39]. Independence, separation, sufficiency, subgroup, and causal fairness metrics must therefore be interpreted relative to the task, protected attributes, data distribution, uncertainty, calibration, and domain constraints. Emerging metrics such as UCerF and domain-specific LLM audit frameworks may complement established metrics, but remain literature-supported and context-specific [41, 42]. Transparency and XAI criteria are mainly protocol-based: faithfulness, robustness, and randomisation checks remain central, while training-data attribution or influence-based analysis may provide supplementary evidence where feasible [44]. Audit trails and contestability are treated separately as accountability controls. Accountability is anchored in the availability, completeness, quality, and traceability of relevant evidence artefacts and governance mechanisms; presence alone is not sufficient to establish effective accountability; this pillar gains new clarity from procurement directives such as OMB M-26-04 and standards such as ISO/IEC 42001 [38, 40]. These sources illustrate how documentation artefacts, evaluation evidence, acceptable-use information and feedback mechanisms are becoming part of accountability and supply-chain governance. EXCALIBUR must therefore maintain a clear traceability chain from regulatory obligation or literature evidence to metric, threshold, interpretation criterion and evidence artefact.

7 EXCALIBUR Taxonomy and Requirements

Section 6 identified which metrics, controls, thresholds, and evidence artefacts can be justified through the five shortlisted instruments and the scientific literature corpus. The purpose of this section is to consolidate those outputs into an EXCALIBUR-oriented taxonomy. The taxonomy does not introduce new metrics or new thresholds. Instead, it reorganises the evidence of Section 6 into a practical requirements structure that can support the design of the EXCALIBUR framework, the LLM-based explainability and fairness components, and the later toolset.

The taxonomy follows the structure: Principle -> Sub-dimension -> Requirement -> Metric / Control -> Threshold / Criterion -> Evidence artefact. This structure is intended to make each EXCALIBUR requirement traceable. A requirement should not appear as an isolated design preference; it should be connected to a principle, a sub-dimension, a measurable or auditable element, an interpretation rule where available, and an artefact that can be stored, inspected, or reused.

The taxonomy is not a final implementation specification. It is a requirements-level bridge between D2.1 and the later technical work packages. WP3 may use it to define what kinds of explanation and fairness outputs should be generated by the LLM-based components. WP4 may use it to define what the platform should display, log, and export. WP5 may use it to structure in-lab and in-the-wild validation, especially where non-expert stakeholders are asked to assess the usefulness, trustworthiness, and accessibility of EXCALIBUR outputs.

7.1 Taxonomy construction logic

Section 7 consolidates the evidence-based mapping of Section 6 into a hierarchical taxonomy of trustworthy AI requirements. The taxonomy is built using three elements. First, it takes the three D2.1 pillars — Fairness, Transparency/XAI and Accountability — as the top-level categories. These pillars correspond to the main failure modes identified earlier in the deliverable: biased outcomes, opaque model behaviour and insufficient responsibility or auditability. Since 2024, new failure modes have emerged in connection with large language models and generative AI, such as hallucination, toxicity and ideological bias. The taxonomy incorporates these dimensions within the existing pillars: hallucination and toxicity are treated as XAI reliability and safety concerns [43], while U.S. procurement guidance such as OMB M-26-04 is treated only as a supplementary example of documentation and vendor-disclosure practices for LLMs, not as a substantive fairness or non-discrimination framework for EXCALIBUR.

Second, it uses the metrics-and-thresholds mapping of Section 6. Fairness requirements are derived from the metric families operationalised in Table 6: independence, separation, sufficiency, subgroup fairness, calibration, and causal or counterfactual fairness. Supplementary LLM-oriented considerations, including uncertainty-aware fairness and domain-specific bias-evaluation frameworks [41, 42], are treated as optional extensions rather than as core Table 6 metrics. Transparency/XAI requirements are derived from Table 7 and cover explanation-method selection, faithfulness, robustness, randomisation sanity checks, concept-based explanations, human-centred evaluation, and stakeholder-facing transparency. Where feasible, influence functions may provide supplementary training-data-attribution evidence [44]. Audit-trail completeness and contestability are classified under Accountability. Accountability requirements are derived from Table 8 and include technical documentation, logs, model cards [15], datasheets [16], audit frameworks [32, 34], process controls [33], responsibility allocation, monitoring records, and supporting governance or risk-management evidence.

Third, it uses EXCALIBUR's methodological orientation. EXCALIBUR does not only calculate technical metrics; it aims to translate them into outputs that non-experts can understand. Therefore, each taxonomy row includes not only the metric or control, but also the evidence artefact that should be produced by the framework or platform. To address new regulatory requirements, where relevant to an LLM procurement context, the taxonomy may also record the availability of model or system cards, acceptable-use policies, end-user resources, feedback mechanisms and additional evaluation evidence [40]. For LLMs, the taxonomy captures hallucination and safety metrics and notes whether these were evaluated using domain-specific benchmarks [43]. This is the part that makes the taxonomy actionable: every requirement should leave an auditable trace.

7.2 EXCALIBUR taxonomy of requirements

Table 9 consolidates the EXCALIBUR-oriented requirements derived from the mappings in Tables 6–8 into a single traceable structure linking each pillar and sub-dimension to a requirement, metric or control, threshold or interpretation criterion, and evidence artefact. The table shows that the three pillars require different forms of operational evidence: Fairness is primarily context- and metric-family dependent; Transparency/XAI combines technical explanation-quality evaluation with stakeholder-facing communication; and Accountability relies mainly on documentation, provenance, auditability, monitoring, and responsibility allocation. Table 9 therefore serves as a

requirements-level bridge to subsequent EXCALIBUR activities rather than as a legal-compliance checklist or final implementation specification.

Principle	Sub-dimension	Requirement	Metric / Control	Threshold / Criterion	Evidence artefact
Fairness	Protected-attribute and task definition	The system shall explicitly define the protected attributes, target outcome, decision task, and affected groups before calculating fairness metrics.	Protected-attribute inventory; task description; decision-output type	No numerical threshold. Mandatory contextual precondition for any fairness assessment.	Fairness assessment setup sheet; dataset documentation; protected-attribute record
Fairness	Independence / group parity	The system shall report whether predicted positive outcomes differ across protected groups where the task involves binary or selection-like decisions.	Statistical Parity Difference; Demographic Parity; Disparate Impact Ratio	For employment-selection systems within the scope of NYC Local Law 144, apply the DCWP impact-ratio calculation and report the results separately for the required sex, race/ethnicity, and intersectional categories [39]. Interpretation must remain linked to the decision context and the reported group-level evidence.	Group fairness report; metric values per group; threshold-source note
Fairness	Separation / error-rate parity	The system shall report whether error rates differ across protected groups when ground-truth labels are available.	Equalised Odds; Equal Opportunity; FPR / FNR balance	No universal threshold. Lower disparity is preferred; interpretation depends on task risk and selected worldview.	Error-rate fairness report; confusion matrices per group; worldview statement
Fairness	Sufficiency / calibration	The system shall report whether predicted scores have comparable meaning across protected groups.	Predictive Parity; Calibration within Groups; subgroup calibration reporting	No universal threshold. Report as diagnostic evidence, not as standalone compliance pass/fail.	Calibration report; reliability plots; model evaluation appendix
Fairness	Subgroup and intersectional fairness	The system shall check whether apparently fair aggregate results hide unfairness in intersectional or smaller subgroups.	γ -Statistical Parity Subgroup Fairness; γ -False Positive Subgroup Fairness; intersectional impact ratios	No established universal threshold. The cited study explored γ values from 0 to 0.029 in a specific experiment. Any EXCALIBUR alert level must be separately pre-registered, justified, and validated; 0.03 is not a literature-derived cut-off.	Subgroup fairness audit; subgroup definition record; alert log
Fairness	Causal and counterfactual fairness	Where causal assumptions are available, the system may assess whether protected attributes causally affect outcomes through impermissible paths.	Counterfactual Fairness; Path-specific Counterfactual Fairness; Direct Counterfactual Fairness	No universal threshold. Requires explicit causal graph or structural assumptions.	Causal-assumption sheet; SCM/DAG documentation; causal fairness report
Fairness	Fairness limitation reporting	The system shall report incompatibilities between fairness metric families instead of collapsing fairness into a single score.	WAE / WYSIWYG worldview statement; incompatibility note	Required whenever multiple fairness families are reported.	Fairness interpretation note; user-facing explanation of metric trade-offs
Transparency / XAI	Explanation method selection	The system shall document which XAI method is used and why it is suitable for the model, data type, and stakeholder.	SHAP; LIME; Integrated Gradients; Grad-CAM; TCAV; counterfactual explanations; other selected methods	No source-defined method threshold. Method must be justified and evaluated.	XAI method card; explanation configuration record
Transparency / XAI	Explanation faithfulness	The system shall evaluate whether explanations reflect model behaviour rather than only producing plausible-looking narratives.	Infidelity; ROAR; literature-supported faithfulness evaluation protocols documented in Table 7	Use metric-specific or relative interpretation: lower Infidelity is preferred; under a valid feature-removal protocol, stronger degradation provides evidence that highly ranked features carry predictive information; where ROAR is used, comparison against an appropriate random baseline should be reported. No universal numerical acceptance threshold is established.	Faithfulness evaluation report; benchmark curves; baseline comparison
Transparency / XAI	Explanation robustness	The system shall evaluate whether explanations are stable under small input perturbations or controlled changes.	Max-Sensitivity [27]	Lower is better; no universal cut-off. Must be interpreted jointly with faithfulness.	Robustness report; perturbation protocol; sensitivity values
Transparency / XAI	Sanity checks for explanations	The system shall run model and/or data randomisation diagnostics before using an attribution method as primary evidence.	Model Parameter Randomisation Test; Data Randomisation Test	Explanations should change under controlled parameter or label randomisation. The similarity metric and acceptance rule must be pre-specified; no universal PASS/FAIL cut-off is defined by [30].	Sanity-check report; method acceptance/deprecation log

Principle	Sub-dimension	Requirement	Metric / Control	Threshold / Criterion	Evidence artefact
Transparency / XAI	Concept-based explanation	Where human-understandable concepts are available, the system may test whether concepts significantly influence model outputs.	TCAV score; CAV validation quality	Statistical significance testing against no-concept influence; protocol depends on the TCAV setup.	Concept library; CAV validation report; TCAV explanation report
Transparency / XAI	Human-centred explanation evaluation	The system shall assess whether explanations are understandable, useful, sufficiently detailed, and appropriate for non-expert stakeholders.	Explanation Satisfaction Scale; Explanation Goodness Checklist; usability / trust / mental-model measures	No universal pass/fail threshold. Use comparative interpretation across explanation variants and stakeholder groups.	User-study report; ESS results; stakeholder feedback summary
Transparency / XAI	Stakeholder-facing communication	The system shall translate technical metrics and explanations into user-facing outputs that state capabilities, limitations, and uncertainty.	Explanation summary; limitations note; uncertainty / confidence communication	No numerical threshold. Must include method, scope, limitations, and intended audience.	User-facing explanation report; platform export; explanation log
Accountability	Technical documentation	The system shall maintain technical documentation that records model purpose, data, methods, performance, fairness metrics, XAI evidence, and limitations.	Technical documentation package; Annex IV-style evidence mapping	Mandatory presence for high-risk AI Act-style evidence; content completeness criterion.	Technical documentation file; EXCALIBUR audit package
Accountability	Dataset documentation	The system shall document dataset origin, composition, preprocessing, sensitive attributes, limitations, and recommended uses.	Datasheets for datasets; data provenance record	Presence and completeness; no universal numerical score.	Dataset datasheet; preprocessing log; data-governance record
Accountability	Model documentation	The system shall document model architecture, intended use, training process, evaluation results, fairness behaviour, and known limitations.	Model card; model registry entry	Presence and completeness; no universal numerical score.	Model card; model-version record; evaluation summary
Accountability	Audit trail	The system shall log every fairness and XAI assessment with model version, dataset version, protected attribute, metric or control, metric/control provenance, threshold status/source, and responsible actor.	Audit log; run ID; model version; dataset version; metric configuration; SMACTR-style audit trail and ADHF [32]; System Provenance and Logging PM 2.1.3 [33]; per-layer audit trail [34]	Mandatory for traceability. Each assessment must be reproducible from stored metadata.	Machine-readable audit log; experiment-tracking record
Accountability	Threshold traceability	The system shall record separately the provenance of each metric or control and the status or source of each threshold or interpretation criterion.	Metric/control provenance field; threshold-status/source field	Required for every reported metric or control. Metric/control provenance should distinguish instrument-defined, literature-supported, and project-defined elements; threshold status should distinguish source-defined, literature-supported, project-defined, and no-established-threshold cases, as applicable.	Traceability register; instrument/publication-to-requirement mapping
Accountability	Post-deployment monitoring	The system shall define which fairness, XAI, and performance indicators must be monitored after deployment or during repeated experiments.	Monitoring plan; drift checks; fairness re-audit; explanation-quality re-check; continuous ex-post application-layer monitoring [34]; Incident Reporting and Response PM 3.1.1 [33]	No universal frequency. Frequency should be risk- and context-dependent.	Monitoring plan; periodic audit report; drift / re-evaluation log
Accountability	Responsibility allocation	The system shall identify who is responsible for data preparation, model training, metric selection, explanation generation, threshold interpretation, and user-facing reporting.	Responsibility table; provider/deployer/developer/auditor role mapping; Roles and Responsibilities PM 1.1.1 and RAI Governance Committee PM 1.1.2 [33]; role mapping across the development organisation [34]	Mandatory role assignment for accountable use.	Responsibility table; governance record; review sign-off
Accountability	LLM-specific audit	Where the framework uses LLM-based components for explanation generation, the system shall apply LLM-specific audit layers in addition to the standard model audit, with continuous ex-post evaluation at the application level.	Three-layered audit (governance / model / application) [34]; application-level bias, safety, robustness, and truthfulness evaluation using explicitly documented benchmarks selected in WP3/WP5	Coverage of all three audit layers; outputs of upper layers inform lower layers; application-level evaluation combines ex-ante assessment with continuous ex-post monitoring. Benchmark-specific interpretation criteria must be documented where such benchmarks are adopted.	LLM audit report; application-layer monitoring log; benchmark and evaluation record

Table 9 — EXCALIBUR taxonomy of trustworthy AI requirements.

The taxonomy has three important implications for EXCALIBUR. First, the framework should not treat metrics as isolated outputs. A fairness metric without a protected attribute, task description, worldview statement, threshold-source note and evidence artefact is incomplete. Similarly, an explanation without a method card, faithfulness or sanity-check evidence, provenance of training data and target-audience description is not sufficient as a transparency output. For generative or LLM-based systems, fairness reports may additionally include uncertainty-aware measures such as UCerF and domain-specific bias-evaluation evidence where these are suitable for the task [41, 42]. XAI reports should document the selected method, faithfulness and robustness evidence, sanity-check results, and user-study findings; where feasible, they may also include training-data attribution or influence-based evidence [44]. Audit-trail completeness and contestability should be reported under Accountability. Accountability reports must list model cards, datasheets, evaluation artefacts, acceptable use policies and feedback mechanisms, along with references to relevant procurement or management-system standards [38, 40].

Second, the taxonomy shows that threshold evidence is uneven across pillars. Fairness and XAI rely heavily on literature-supported and relative criteria, while accountability relies more on source-defined documentation and process criteria. This means that EXCALIBUR should distinguish between metric value, interpretation criterion and evidence completeness. A low or high metric value is not automatically a compliance statement. Emerging evaluation frameworks emphasise that fairness and explainability must also consider uncertainty, calibration and domain-specific context; therefore, the taxonomy records these factors as metadata rather than forcing them into a single threshold [41, 42].

Third, the taxonomy creates the minimum evidence package that should accompany an EXCALIBUR report. Each generated fairness report or explanation should be accompanied by an audit trail showing: dataset version, model version, method, protected attribute, metric family, metric/control provenance, threshold status or source, limitations, responsible actor and — where applicable — training-data summary, model card reference, evaluation artefact ID, acceptable use policy reference and feedback mechanism. In addition, LLM-based reports should include hallucination and safety evaluation results, domain-specific benchmark references, provenance information, and documentation of any feedback or review mechanism used to challenge unreliable outputs [40, 43].

7.3 Indicative scenario 1 - Wearable well-being prediction

This scenario illustrates how the D2.1 taxonomy could be applied in a wearable-domain validation setting. It is not intended to prescribe the final EXCALIBUR implementation, metric set, or acceptance criteria. The concrete methods, metrics, interpretation rules, and evaluation procedures will be selected and validated in the relevant downstream work packages.

The first indicative scenario follows the EXCALIBUR validation domain: a wearable-data model predicts a health- or well-being-related state, such as stress level, sleep quality or risk of reduced well-being. The model uses time-series or aggregated features from wearable sensors and produces either a binary classification, a risk score or a recommendation. If the model includes an LLM component, the evaluation may additionally include uncertainty-aware fairness measures such as UCerF, alongside established fairness metrics and domain-specific bias checks, where these are suitable for the task [41, 42].

Before any metric is calculated, the system records the task, output type, dataset version, model version and protected or sensitive attributes available for fairness analysis. The dataset is accompanied by a datasheet describing composition, preprocessing, missingness and known limitations. For the Fairness pillar, the system reports group-level metrics only where the output type supports them. If the model produces a binary risk label, the report may include Statistical Parity Difference, Disparate Impact Ratio, Equal Opportunity, Equalised Odds and subgroup fairness diagnostics. If the model produces a continuous risk score, calibration and score-distribution diagnostics become more relevant. When an LLM or generative model is involved, fairness reports may additionally

include UCerF [41] or similar uncertainty-aware measures, together with domain-specific bias assessments, provided that their experimental status and assumptions are documented [41, 42].

For the Transparency/XAI pillar, the system generates technical explanations using methods suitable for the model and data type. For tabular aggregates, SHAP or LIME may be used; for temporal models, method selection must be justified more carefully because some standard XAI methods may not transfer cleanly to time-series data. Where feasible and methodologically justified, explanations may be complemented by training-data attribution or influence-based evidence [44]. Such evidence is supplementary and should not be treated as a universal requirement for all model or data types. If an LLM is used to generate human-readable explanations, the evaluation should consider the explanation's faithfulness, robustness, factual consistency, and unsupported-claim or hallucination behaviour using the protocol selected in the relevant downstream work [43]. The explanation is not directly passed to the user as final evidence. It is first evaluated through faithfulness, robustness and sanity-check criteria where feasible. The LLM-based component then uses the validated explanation outputs, the model prediction and the fairness results to generate a non-expert explanation.

For the Accountability pillar, the platform stores the full evidence package: dataset datasheet, model card, fairness report, XAI method card, explanation-quality report, threshold-source register, audit log and procurement artefacts if relevant (e.g. evaluation artefacts or acceptable use policy). The user-facing output may state, for example: The model prediction was mainly influenced by sleep regularity and activity variability; however, fairness checks showed higher false-negative rates for one age group and the explanation passed faithfulness and robustness checks but displayed a non-negligible hallucination rate. This should be reviewed before deployment. The evidence artefacts allow a researcher or auditor to trace that statement back to the underlying metrics and methods.

This scenario demonstrates the main role of the taxonomy: it turns EXCALIBUR from a system that simply produces explanations and fairness metrics into a system that produces traceable, documented and interpretable evidence. The updated scenario shows how emerging fairness and XAI metrics, uncertainty-aware evaluation and supplementary transparency and accountability evidence can be integrated into the EXCALIBUR framework.

7.4 Indicative scenario 2 - Automated selection or ranking system

This second scenario is a comparative and illustrative application of the taxonomy rather than a committed EXCALIBUR pilot. It demonstrates how the same requirements structure could be applied to a selection or ranking context in which source-defined audit and disclosure rules, such as NYC Local Law 144, provide a useful operational comparator.

The second indicative scenario concerns an automated selection or ranking system, such as an employment-related screening tool or another decision-support system that ranks individuals for access to an opportunity. This scenario is useful because it closely follows the operational bias-audit logic and worked examples of NYC Local Law 144 and the DCWP Final Rule [39], even though it is not the primary EXCALIBUR pilot domain.

The system first defines the decision task: for example, whether the AI system recommends, ranks or screens candidates. The system records the protected attributes used for the audit and the groups for which selection rates will be reported. Since this is a selection-like task, selection-rate and impact-ratio reporting become directly relevant.

For the Fairness pillar, the system reports selection rates and impact ratios for each required group and intersectional category. The platform reports selection rates and impact ratios for each group and intersectional subgroup and highlights the largest relative disparities for further review. The report records the calculation source, group definitions, sample sizes, dataset version, and contextual limitations [39]. Error-rate metrics such as Equal Opportunity or Equalised Odds may also be reported if ground-truth outcome labels are available, but the report

must explain that these metrics belong to a different fairness family and may conflict with demographic-parity metrics. If the system uses a generative model (e.g. to suggest interview questions or to rank CV narratives), fairness assessments may additionally include uncertainty-aware metrics and domain-specific bias checks where these are suitable for the task; for example, whether the model produces biased summaries for certain demographic groups [41].

For the Transparency/XAI pillar, the system documents which characteristics or features influenced the recommendation and provides a stakeholder-facing explanation of the model's capabilities and limitations. If feature-attribution methods are used, they must be supported by method documentation and, where possible, sanity-check evidence. Where feasible, feature-attribution outputs may be complemented by training-data attribution or influence-based evidence [44]. If an LLM generates the user-facing explanation, the output should also be evaluated for factual consistency and unsupported reasoning using the evaluation protocol selected for WP3 and WP5 [43]. The user-facing report should not only say which features were important, but also whether the explanation method passed basic reliability checks.

For the Accountability pillar, the system produces an audit package comprising the applicable bias-audit report, model card, dataset datasheet, threshold-source note, public-facing summary, and internal audit log. Procurement-specific artefacts, such as evaluation artefacts or acceptable-use policies, are added only where a relevant procurement context applies. If the system is used in a NYC LL 144-like context, the taxonomy also records the notice and disclosure requirements as process-side transparency and accountability evidence. If the deployment concerns an LLM procurement by a U.S. federal agency within the scope of OMB M-26-04, the procurement record should additionally include the applicable vendor-transparency documentation [40]. The final report therefore connects selection-rate metrics, threshold interpretation, documentation and disclosure into a single traceable evidence chain.

This scenario shows why the taxonomy is necessary. Without it, the system might report a single impact-ratio value and treat it as a complete fairness audit. With the taxonomy, the same value becomes part of a documented chain: protected attribute → selection task → metric → threshold source → interpretation limitation → audit report → responsible actor → procurement evidence. The updated scenario illustrates how generative AI and procurement requirements introduce new evidence artefacts and fairness metrics into the audit chain.

7.5 Requirements carried forward to WP3, WP4 and WP5

For WP3, the LLM-based explainability and fairness methodologies should consume structured evidence rather than raw metric values alone. The LLM should receive the metric name, value, protected attribute, threshold source, limitation, model version, dataset version, explanation-quality evidence, and audit-trail references. Where available and technically justified, the structured input may also include training-data attribution or influence-based evidence [44]. For LLM-generated outputs, it should additionally include the factuality, safety, provenance, and feedback evidence selected for the evaluation protocol [40, 43]. This reduces the risk that the LLM generates over-confident or unsupported explanations. WP3 should also implement evaluation protocols that combine automated metrics with human judgement to assess LLM outputs, as recommended in recent surveys [43].

For WP4, the platform should implement evidence artefacts as first-class objects. The toolset should be able to export fairness reports, XAI reports, model cards, datasheets, audit logs, threshold registers, model-documentation templates, evaluation artefacts, acceptable use policies and user-feedback mechanisms. The platform should also distinguish visually between source-defined, literature-supported, project-defined and non-established thresholds. Where relevant, WP4 may support uploading or exporting model cards, evaluation artefacts, acceptable-use notes, and feedback-mechanism records, so that EXCALIBUR outputs remain compatible with emerging documentation and procurement practices without treating any single external procurement framework as binding.

For WP5, the validation process should evaluate not only whether users understand the explanation, but also whether the explanation is supported by technical evidence. Human-centred evaluation should therefore be paired with faithfulness, robustness, hallucination and auditability checks. This is especially important in the wearable domain, where non-expert users may rely on explanations related to sensitive health or well-being data. WP5 should also test whether LLM-based explanations help users detect hallucinations or model bias and whether additional information such as training-data provenance improves user trust [43].

Figure 2 situates D2.1 within the broader EXCALIBUR work-package and deliverable flow. D2.1 provides the Trustworthy AI principles and technical requirements mapping, while D2.2 provides the complementary human-centric and ethics requirements of WP2. Together, these WP2 outputs establish a requirements baseline that informs the explainability and fairness methodologies of WP3, the framework and toolset implementation of WP4, and the technical and human-centred evaluation activities of WP5. The relationships shown are functional rather than a strict representation of deliverable submission chronology, since development and validation activities overlap across the project.

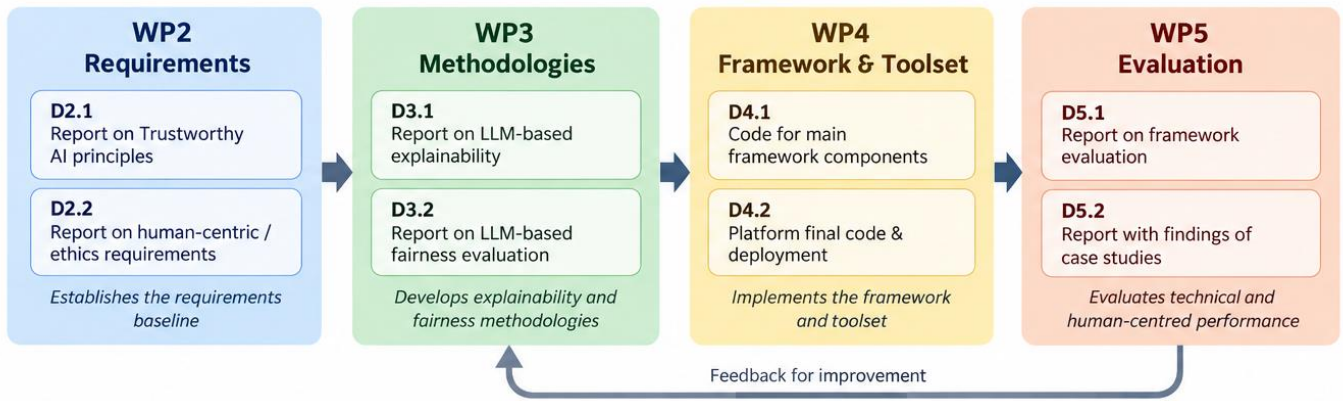


Figure 2. Functional position of D2.1 within the EXCALIBUR deliverable flow. WP2 establishes the requirements baseline, WP3 develops the explainability and fairness methodologies, WP4 implements the framework and toolset, and WP5 evaluates the resulting methods and platform. Arrows represent functional dependencies and feedback rather than strict deliverable submission chronology.

Table 10 summarises the requirements-level hand-off from D2.1 to WP3, WP4, and WP5. It separates methodology definition in WP3, framework and toolset implementation in WP4, and validation and verification in WP5, while associating each requirement area with the evidence expected to demonstrate its implementation. The key distinction is that the same D2.1 requirement is carried forward differently across the three work packages: WP3 defines the relevant methodologies, WP4 operationalises them through framework and platform functionality, and WP5 evaluates their technical and human-centred performance.

Requirement area	WP3 — methodology definition	WP4 — framework/toolset implementation	WP5 — validation / verification	Expected evidence
Fairness assessment	Select and justify applicable fairness metric families, subgroup/causal methods, and interpretation logic for the relevant model and task.	Implement calculation, configuration, storage, presentation, and export of fairness outputs and associated metadata.	Verify metric behaviour, contextual appropriateness, subgroup effects, and interpretation with the selected experimental settings.	Fairness report; metric configuration; protected-attribute/task record; validation results.
Transparency/XAI methods and technical evaluation	Develop/select explanation methodologies and define faithfulness, robustness, sanity-check, and other applicable evaluation protocols.	Integrate explanation methods and evaluation outputs into the framework; store method/configuration information and expose results through the toolset.	Validate technical explanation quality and compare methods or configurations where applicable.	XAI method record; faithfulness/robustness report; sanity-check evidence; validation results.
Human-centred explanation and communication	Define the information and structured evidence that should support stakeholder-facing explanations.	Present explanations, limitations, provenance, and uncertainty through stakeholder-facing interfaces and exports.	Evaluate understandability, usefulness, trust calibration, and other human-centred criteria with relevant stakeholders.	User-facing explanation; user-study protocol; stakeholder feedback and evaluation results.
Accountability and evidence traceability	Ensure methodology outputs contain sufficient provenance, limitations, model/dataset identifiers, and evidence references.	Implement persistent storage, versioning, audit logs, evidence packages, model/dataset documentation, and export functionality.	Verify that outputs are traceable, sufficiently complete, and reproducible from the retained evidence.	Audit log; model card; datasheet; evidence package; traceability check.
Metric/control provenance and threshold status	Label operational elements and interpretation rules according to the provenance/status scheme defined in D2.1.	Implement provenance and threshold-status fields and make the distinction visible in the toolset.	Verify that experimental/project-defined criteria are not presented as externally mandated or universal thresholds.	Provenance/threshold register; UI/export evidence; validation record.
Monitoring and LLM-specific evaluation	Define applicable LLM/application-level evaluation and monitoring methods where LLM-based components are used.	Support logging, versioning, repeated evaluation, and monitoring of relevant LLM/framework outputs.	Perform repeated technical and human-centred evaluation, including ex-post assessment where required by the selected experimental protocol.	LLM/application audit report; monitoring log; repeated-evaluation results.

Table 10 — Requirements-level hand-off from D2.1 to WP3, WP4, and WP5.

Overall, the Section 7 taxonomy converts the evidence-based mapping of Section 6 into a practical EXCALIBUR requirements structure. It does not close the open methodological gaps identified in the previous sections, but it makes them explicit and traceable. The updated taxonomy accommodates new fairness metrics and audit frameworks for generative models, enterprise-level explainability features, procurement-driven accountability artefacts and safety metrics for LLM outputs. By integrating these developments into the EXCALIBUR framework, WP3, WP4 and WP5 can remain aligned with evolving trustworthy AI practice and regulation.

8 Conclusions

D2.1 provided the initial mapping foundation for the EXCALIBUR work on Trustworthy AI principles. The deliverable focused on three pillars - Transparency, Fairness, and Accountability - and examined how these principles are expressed across regulatory instruments, standards, governance frameworks, and scientific literature. Its main contribution is not to propose a new regulatory framework, but to translate existing trustworthy AI expectations into a structured set of technical and evidence-based elements that can support the later EXCALIBUR framework, toolset, and validation activities.

The deliverable achieved three main outcomes. First, it developed a broad mapping of regulations, standards, frameworks, and guidelines relevant to trustworthy AI, identifying which instruments address Transparency, Fairness, and Accountability and at what level of operational detail. Second, it mapped the selected instruments to operational metrics, controls, documentation artefacts, and threshold sources, distinguishing between instrument-defined requirements, literature-supported operational elements and interpretation criteria, project-defined experimental rules, and cases in which no established threshold exists. Third, it consolidated these outputs into an EXCALIBUR-oriented taxonomy of requirements, linking each principle to sub-dimensions, requirements, metrics or controls, threshold or interpretation criteria, and evidence artefacts.

8.1 General conclusions from the regulatory, standards and literature landscape

A central conclusion of D2.1 is that the Trustworthy AI landscape has matured significantly at the level of principles, obligations, and governance structures, but remains uneven at the level of technical operationalisation. Regulations and standards increasingly require AI systems to be transparent, fair, accountable, documented, monitored, and auditable. However, most instruments do not define the exact metric, method, or numerical threshold that should be used to demonstrate that these requirements have been met.

This gap is especially visible in fairness and explainability. Fairness is supported by a rich scientific literature, but the available metrics encode different assumptions and are not mutually compatible in all settings. Independence-based metrics, separation-based metrics, sufficiency-based metrics, subgroup fairness metrics, and causal fairness definitions answer different questions and may lead to different conclusions for the same model. Therefore, fairness cannot be reduced to one universal score or one universal threshold. Where numerical material exists, such as the worked impact-ratio calculations in the DCWP Final Rule [39] or experimental γ values reported in subgroup-fairness studies [14], it must be interpreted according to its source, methodological purpose, and use context.

The same applies to transparency and XAI. Regulatory instruments require systems to be sufficiently transparent, documented, and understandable to relevant actors, but they do not prescribe a single XAI method or a universal explanation-quality cut-off. The literature provides useful evaluation protocols, such as faithfulness measures, robustness measures, randomisation sanity checks, ROAR-style benchmarks, concept-based testing, and human-centred explanation evaluation. Yet these are mostly relative, protocol-based, or task-specific criteria. A method such as SHAP, LIME, TCAV, or Integrated Gradients cannot be treated as trustworthy by default; it must be evaluated and documented in relation to the model, data type, use case, and stakeholder.

Accountability is different. It is less often expressed through mathematical metrics and more often through documentation, logs, audit trails, risk management records, post-market monitoring, model cards, datasheets, audit reports, and responsibility allocation. This makes accountability the most concrete pillar in terms of evidence artefacts, even when no numerical threshold exists. The main challenge is therefore not only to calculate metrics, but to ensure that every metric, explanation, threshold, and decision rule is traceable to the originating instrument, scientific publication, or explicitly documented project-defined decision and stored in a form that can be reviewed by researchers, auditors, deployers, or affected stakeholders.

Overall, D2.1 shows that the current trustworthy AI landscape is not lacking principles. It is lacking consistent bridges between legal or normative requirements and technical evidence. This is the gap that the deliverable addresses by

separating what is directly required by regulations or standards, what is supported by literature, what must be defined by the project for experimentation, and what remains an open methodological issue.

8.2 EXCALIBUR-specific conclusions and next steps

D2.1 provides EXCALIBUR with a traceable requirements baseline for the design of its LLM-based explainability and fairness methodologies and for the accountability evidence layer surrounding them. Its contribution is not a legal compliance determination, but a structured translation of regulatory, standardisation, and literature-based expectations into candidate metrics, evaluation protocols, process controls, interpretation criteria, and reviewable evidence artefacts. The taxonomy in Section 7, together with Tables 6–8, the requirements hand-off in Table 10 and Figure 2, and the consolidated catalogues in Appendices A and B, records both the operational elements selected for EXCALIBUR and the broader body of evidence from which those selections were made.

For WP3, this baseline specifies the evidence that should inform—and, where appropriate, serve as structured input to—the LLM-based explainability and fairness methodologies. This evidence includes the task and stakeholder context; protected attributes; the selected fairness metric family and associated worldview; metric values and uncertainty; metric/control provenance and threshold status or source; model and dataset versions; explanation-faithfulness, robustness, and sanity-check results; and known methodological limitations. LLM-generated explanations and fairness reports should therefore be treated as evidence-grounded summaries rather than as independent sources of technical or legal conclusions.

For WP4, the taxonomy translates into a minimum evidence package that the EXCALIBUR toolset should be able to produce, store, display, and export. This package should include fairness and XAI evaluation reports, method and configuration records, model cards, dataset datasheets, threshold-source registers, audit logs, monitoring records, and user-facing summaries. The toolset should clearly distinguish instrument-defined obligations, literature-supported criteria, project-defined experimental rules, and cases in which no established threshold exists. This distinction is necessary to prevent an experimental warning or diagnostic result from being interpreted as a statement of legal compliance.

For WP5, the same structure provides a basis for technical and human-centred validation. Evaluation should assess not only readability and perceived usefulness, but also explanation faithfulness, robustness, traceability, uncertainty communication, and the ability of non-expert users to recognise limitations or unreliable outputs. These requirements are particularly important in the wearable and well-being domain, where sensitive data, temporal dependencies, and subgroup effects may make both explanation and fairness assessment more difficult.

The immediate next steps are to translate the requirements in Section 7 into implementation specifications for WP3 and WP4; address the remaining methodological gaps concerning time-series and wearable data, LLM-based explanation and fairness evaluation, human-centred validation, standardised quality and completeness criteria for accountability evidence, machine-readable evidence schemas, and continuous audit protocols; and explicitly define, version, and log every project-defined threshold or interpretation rule used during the in-lab and in-the-wild studies. Such rules should remain explicitly provisional and should not be presented as universal, regulatory, or legally binding thresholds.

D2.1 therefore establishes the conceptual, technical, and evidentiary basis for the next phases of EXCALIBUR. By connecting each requirement to its originating instrument or scientific publication, operational measure, interpretation criterion, limitation, and evidence artefact, it enables the project to move from high-level trustworthy-AI principles towards auditable, technically grounded, and human-centred assessment of explainability, fairness, and accountability.

Appendix A — Fairness metric definitions

This appendix gives the formal definition of each group-fairness criterion mapped in Table 6 and, where available, identifies a corresponding implementation or diagnostic in the AIF360 toolkit [46]. Table A.1 summarises the notation used in the formal definitions below. Several combinations of the criteria below cannot generally be satisfied simultaneously when group base rates differ and predictions are imperfect [17, 18, 23, 25, 45]. Metric selection must therefore be justified in relation to the use case, the selected fairness worldview, and the relative costs of different errors. Multiple metrics may be reported as complementary diagnostics, provided that their conceptual differences and incompatibilities are stated explicitly.

Notation	Description
A	Protected or sensitive attribute.
a, b	Two values or comparison groups of the protected attribute A.
Y	Ground-truth label or observed outcome.
$\hat{Y}$	Model decision or predicted label.
S	Continuous risk or prediction score.
s, y	Realised values of the score S and the ground-truth label Y, respectively.
$P(\cdot)$	Probability of the event inside the parentheses.
$E[\cdot]$	Expectation (mean) of the quantity inside the brackets.
p_a, p_b	Group-specific positive-decision rates: $P(\hat{Y} = 1 \mid A = a)$ and $P(\hat{Y} = 1 \mid A = b)$.
SPD	Statistical Parity Difference, defined as $p_a - p_b$.
$DI_{a,b}$	Generic two-group disparate-impact ratio, defined as p_a / p_b .
G	Set of categories compared in a category-wise audit.
g, h	Category indices belonging to G.
r_g	Selection or scoring rate for category g, as applicable.
IR_g	Category-wise impact ratio: r_g divided by the highest corresponding rate among the compared categories.
TPR / FPR	True-positive rate / false-positive rate.
PPV / NPV	Positive predictive value / negative predictive value.

Table A.1 — Notation used in Appendix A.

A.1 Demographic parity and related disparity measures

Demographic parity requires equal positive-decision rates across protected groups:

$$P(\hat{Y} = 1 \mid A = a) = P(\hat{Y} = 1 \mid A = b).$$

Let

$$p_a = P(\hat{Y} = 1 \mid A = a), \quad p_b = P(\hat{Y} = 1 \mid A = b).$$

The statistical-parity difference and a generic two-group disparate-impact ratio are then written as:

$$SPD = p_a - p_b$$
$$DI_{a:b} = \frac{p_a}{p_b}$$

where the two quantities defined above are the positive-decision rates of the comparison groups. The ideal values are 0 for SPD and 1 for the two-group DI ratio. For audits under NYC Local Law 144 and the DCWP Final Rule, the operational impact ratio is defined category-wise as:

$$IR_g = \frac{r_g}{\max_{h \in G} r_h}$$

where the numerator is the selection or scoring rate for the category under evaluation, as applicable, and the denominator is the highest corresponding rate among all compared categories. The DCWP Final Rule provides worked examples of this calculation and requires reporting across the specified sex, race/ethnicity, and intersectional categories [39]. Corresponding AIF360 ClassificationMetric methods: `statistical_parity_difference()` and `disparate_impact()`.

A.2 Equalised odds

Equalised odds requires equal true-positive and false-positive rates across protected groups:

$$P(\hat{Y} = 1 \mid Y = y, A = a) = P(\hat{Y} = 1 \mid Y = y, A = b), \quad y \in \{0,1\}.$$

AIF360 diagnostic: `average_odds_difference`. Because a single average may conceal offsetting disparities, the true-positive-rate and false-positive-rate differences should also be reported separately.

A.3 Equal opportunity

Equal opportunity is the positive-class component of equalised odds and requires equal true-positive rates:

$$P(\hat{Y} = 1 \mid Y = 1, A = a) = P(\hat{Y} = 1 \mid Y = 1, A = b).$$

AIF360 implementation: `equal_opportunity_difference`.

A.4 Predictive-value parity

Positive predictive-value parity requires equal precision across protected groups:

$$P(Y = 1 \mid \hat{Y} = 1, A = a) = P(Y = 1 \mid \hat{Y} = 1, A = b).$$

Negative predictive-value parity is the corresponding condition for negative decisions:

$$P(Y = 0 \mid \hat{Y} = 0, A = a) = P(Y = 0 \mid \hat{Y} = 0, A = b).$$

AIF360 implementation: calculated from the per-group confusion-matrix quantities exposed by ClassificationMetric, including `positive_predictive_value` and `negative_predictive_value`.

A.5 Calibration within groups

Calibration within groups requires a probabilistic risk score to have the same empirical meaning within each protected group and to correspond to the observed outcome frequency. For score values s with positive support—or almost everywhere with respect to the score distribution—the condition is:

$$P(Y = 1 \mid S = s, A = a) = P(Y = 1 \mid S = s, A = b) = s.$$

For continuous scores, calibration should be assessed through per-group reliability curves and appropriate calibration-error estimators, rather than by requiring equality at every exact observed score value.

A.6 Balance for the positive and negative classes

Balance for the positive class requires equal mean scores among individuals with positive ground-truth outcomes:

$$E[S \mid Y = 1, A = a] = E[S \mid Y = 1, A = b].$$

Balance for the negative class requires equal mean scores among individuals with negative ground-truth outcomes:

$$E[S \mid Y = 0, A = a] = E[S \mid Y = 0, A = b].$$

Demographic parity, error-rate parity, predictive-value parity, calibration, and class-balance criteria cannot generally all hold simultaneously when group base rates differ and predictions are imperfect [17, 18, 23, 25, 45]. Table 6 should therefore identify the primary fairness criterion for each obligation and may report additional metrics only as complementary diagnostics, with their incompatibilities and interpretation limits stated explicitly.

Appendix B — Consolidated Task 2.1 metrics, evaluation frameworks, and accountability instruments catalogue

This appendix consolidates the metrics, evaluation frameworks, and accountability instruments identified in the original Task 2.1 literature-review corpus (18 papers, P001–P018), together with the supplementary sources added during the final update of the deliverable. Appendix A provides the canonical definitions of the group-fairness criteria operationalised in Table 6. Sections B.1–B.3 document the additional fairness, explainability/XAI, and accountability elements adopted into Tables 6–8 or discussed in the main narrative. Section B.4 records further metrics surveyed in the original corpus but not carried forward. Section B.5 separately identifies supplementary sources [41]–[44], so that later additions are not presented as part of the original P001–P018 corpus. Symbols already defined in Table A.1 retain the same meaning throughout Appendix B. Table B.2 summarises the additional notation introduced in the subgroup-fairness, causal-fairness, XAI-evaluation, and accountability formulations below.

Table B.1 consolidates the metrics, evaluation methods, and accountability instruments identified in the Task 2.1 literature review and records whether each element was carried forward into the main operational tables or retained for reference only. The catalogue shows that Fairness contributes the largest set of formal metrics, Transparency/XAI combines quantitative metrics with procedural and human-centred evaluation protocols, and Accountability is represented primarily through documentation and audit frameworks.

Table B.1 — Consolidated metrics and instruments from the Task 2.1 literature review

Metric / instrument	Category	Source reference	Definition / use location
Statistical parity / disparate impact	Fairness	[20, 24, 45, 46]	Appendix A and Table 6
Equalised odds	Fairness	[20, 24, 45, 46]	Appendix A and Table 6
Equal opportunity	Fairness	[20, 24, 45, 46]	Appendix A and Table 6
Predictive-value parity (PPV / NPV)	Fairness	[20, 23, 24, 45, 46]	Appendix A and Table 6
Calibration within groups	Fairness	[23, 24, 25, 45]	Appendix A and Table 6
Balance for the positive / negative class	Fairness	[23, 45]	Appendix A and Table 6
γ -Statistical Parity subgroup fairness	Fairness	[14]	Table 6
γ -False Positive subgroup fairness	Fairness	[14]	Table 6
Counterfactual fairness	Fairness	[13, 21, 23]	Table 6
Path-specific counterfactual fairness	Fairness	[23]	Table 6
Direct counterfactual fairness	Fairness	[23]	Table 6
Counterfactual equalised odds	Fairness	[25]	Table 6
Principal fairness	Fairness	[25]	Table 6
Conditional demographic disparity	Fairness	[21]	Surveyed only (B.4)
Fairness through unawareness	Fairness	[21]	Surveyed only (B.4)
Fairness through awareness	Fairness	[21]	Surveyed only (B.4)
No unresolved discrimination	Fairness	[21]	Surveyed only (B.4)
No proxy discrimination	Fairness	[21]	Surveyed only (B.4)
Fairness index	Fairness	[21]	Surveyed only (B.4)
Interventional fairness	Fairness	[23]	Surveyed only (B.4)
Equal causal fairness	Fairness	[23]	Surveyed only (B.4)
Conditional principal fairness	Fairness	[25]	Surveyed only (B.4)
Counterfactual predictive parity	Fairness	[25]	Surveyed only (B.4)
Statistical parity for regression	Fairness	[22]	Surveyed only (B.4)
Bounded group loss	Fairness	[22]	Surveyed only (B.4)
Equalising disincentives	Fairness	[24]	Surveyed only (B.4)
Conditional equal opportunity	Fairness	[24]	Surveyed only (B.4)
Bayesian fairness	Fairness	[24]	Surveyed only (B.4)
Generalised entropy index	Fairness	[20, 24]	Surveyed only (B.4)
Theil index	Fairness	[20, 24]	Surveyed only (B.4)
Inframarginality	Fairness	[25]	Surveyed only (B.4)
TCAV / concept sensitivity	Transparency / XAI	[26]	Table 7
Infidelity	Transparency / XAI	[27]	Table 7
Max-Sensitivity	Transparency / XAI	[27]	Table 7
Model Parameter Randomisation Test	Transparency / XAI	[30]	Table 7
Data Randomisation Test	Transparency / XAI	[30]	Table 7
ROAR	Transparency / XAI	[31]	Table 7
Explanation Satisfaction Scale	Transparency / XAI	[29]	Table 7
Quantus six-category framework	Transparency / XAI	[28]	Table 7 and §6.3
SmoothGrad family	Transparency / XAI	[27]	Surveyed only (B.4)
Explanation Goodness Checklist	Transparency / XAI	[29]	Surveyed only (B.4)
Mental-model / trust / curiosity measures	Transparency / XAI	[29]	Surveyed only (B.4)
SMACTR internal-audit framework	Accountability	[32]	Table 8
Datasheets for Datasets	Accountability	[16]	Table 8
Model Cards	Accountability	[15]	Table 8
Responsible AI Metrics Catalogue	Accountability	[33]	Table 8
Three-layered LLM audit blueprint	Accountability	[34]	Table 8 and §6.4
UCerF uncertainty-aware fairness	Fairness	[41]	§6.2, §6.5, and §7.1
Healthcare LLM bias-evaluation framework	Fairness / Accountability	[42]	§6.2, §6.5, and §6.6
LLM evaluation survey	Transparency / XAI	[43]	§6.2–§7.1
Influence functions / training-data attribution	Transparency / Accountability	[44]	§6.3, §6.5, and §7.1

Location legend. “Appendix A” indicates a canonical fairness definition; “Table 6/7/8” indicates that the metric, method, or artefact is carried into the corresponding threshold table; “Narrative” indicates substantive discussion in the specified section; and “Surveyed only (B.4)” indicates that the item was documented in the original literature review but not adopted into a table. Section B.5 identifies supplementary sources added after the original P001–P018 corpus. The source-reference column contains bibliographic references only; appendix or section locations are recorded exclusively in the final column.

Notation	Description
$\mathcal{P}, \Pr_{\mathcal{P}}$	Population distribution and probability evaluated under that distribution.
X, x	Input random vector and a realised input value.
$G, g(X)$	Pre-specified class of subgroup indicators and a subgroup-membership indicator $g: X \rightarrow \{0,1\}$.
D	Classifier or decision function used in the subgroup-fairness definitions.
γ	Subgroup-fairness tolerance parameter; not a universal acceptance threshold.
w_g, d_g, p, p_g	Subgroup prevalence weight, positive-decision-rate disparity, overall positive-decision rate, and subgroup positive-decision rate.
$w_g^{(0)}, d_g^{(0)}, q, q_g$	Negative-label subgroup weight, false-positive-rate disparity, overall false-positive rate, and subgroup false-positive rate.
$\hat{Y}(a), \hat{Y}(a,m)$	Counterfactual model outputs under intervention $A \leftarrow a$ and, where applicable, joint intervention $A \leftarrow a$ and $M \leftarrow m$.
M, Z	Admissible mediating variables and the remaining observed variables in path-specific counterfactual fairness.
$p_a(m,z), p_b(m,z)$	Conditional positive-decision probabilities under the two protected-attribute interventions used in pCFF.
$\text{Desc}(A)$	Set of descendants of A in the assumed causal graph.
$Y(0), Y(1), W, \perp$	Potential outcomes, observed covariates used in conditional principal fairness, and the independence symbol.
C, l, k, m	User-defined concept, model layer, target class, and dimensionality m of the layer- l activation space in the TCAV formulation.
$v_c^{(l)}$	Concept Activation Vector (CAV) for concept C at layer l .
$s(x;C,k,l), X_k, \text{TCAV}(C,k,l)$	Conceptual sensitivity, the set of class- k examples, and the corresponding TCAV score.
$\Phi(f,x), f$	Explanation produced for model f at input x , and the model itself.
I, μ	Perturbation vector and its specified perturbation distribution in the Infidelity metric.
$B(x,r), r, \text{MaxSens}$	Local neighbourhood of x , its radius, and the Max-Sensitivity robustness measure.
$\mathcal{E}, f_{\theta}, f_{\theta}, \mathcal{E}_T(x), \mathcal{E}_R(x)$	Explanation method, trained model, randomised model, and the corresponding explanations in the randomisation tests.
$D_P, \mathcal{T}, f_P$	Training data with permuted labels, training procedure, and the model retrained on the permuted-label data.
t	Fraction of features removed or masked in the ROAR benchmark.
α	Context-dependent parameter: Cronbach’s alpha in human-centred evaluation; $\alpha = 1$ denotes the Theil-index case of the generalised-entropy family in B.4.
c_j, j, J	Binary completion/evidence indicator for accountability process check j , its index, and the total number of applicable checks.

Table B.2 — Additional notation used in Appendix B.

B.1 Fairness metrics and definitions

Subgroup fairness and fairness gerrymandering [14]

Let G be a pre-specified class of subgroup indicators $g: X \rightarrow \{0,1\}$, $\mathcal{P}$ the population distribution, and D the classifier. Statistical-parity subgroup fairness requires

$$w_g d_g \leq \gamma, \quad \forall g \in G,$$

where

$$\begin{aligned} w_g &= \Pr_p[g(X) = 1], \\ d_g &= |p - p_g|, \\ p &= \Pr[D(X) = 1], \quad p_g = \Pr[D(X) = 1 \mid g(X) = 1]. \end{aligned}$$

False-positive subgroup fairness is defined analogously:

$$w_g^{(0)} d_g^{(0)} \leq \gamma, \quad \forall g \in G,$$

where

$$\begin{aligned} w_g^{(0)} &= \Pr_p[g(X) = 1, Y = 0], \\ d_g^{(0)} &= |q - q_g|, \\ q &= \Pr[D(X) = 1 \mid Y = 0], \quad q_g = \Pr[D(X) = 1 \mid g(X) = 1, Y = 0]. \end{aligned}$$

The tolerance γ is not a universal fairness threshold. It depends on the audited subgroup class G , dataset size, sampling uncertainty, and use context. Any EXCALIBUR alert value must therefore be pre-registered, justified, and labelled as project-defined.

Causal and counterfactual fairness [13, 21, 23, 25]

Counterfactual fairness (CF). Let $\hat{Y}(a)$ denote the model output for the same individual and latent state under the intervention $A \leftarrow a$. Counterfactual fairness requires the conditional distribution of the output to remain unchanged when the protected attribute is counterfactually altered [13, 21, 23]:

$$P(\hat{Y}(a) = y \mid X = x, A = a) = P(\hat{Y}(a') = y \mid X = x, A = a)$$

for all admissible a, a', x , and output values y . Restricting a predictor to appropriate non-descendants of A is a sufficient construction in some settings, but it is not the definition itself.

Path-specific counterfactual fairness (pCFF). Let M denote the mediating variables whose causal influence from A is considered admissible, and let Z denote the remaining observed variables. Let $\hat{Y}(a, m)$ denote the model output under the joint intervention $A \leftarrow a$ and $M \leftarrow m$. Define:

$$\begin{aligned} p_a(m, z) &= P(\hat{Y}(a, m) = 1 \mid A = a, M = m, Z = z), \\ p_b(m, z) &= P(\hat{Y}(b, m) = 1 \mid A = b, M = m, Z = z). \end{aligned}$$

The pCFF condition is:

$$p_a(m, z) = p_b(m, z)$$

for all admissible a, b, m , and z . The choice of admissible mediators and all causal assumptions must be documented [23].

Direct counterfactual fairness (dCFF). dCFF is a special path-specific criterion in which mediated effects through the accepted descendants of A are treated as admissible and the direct effect of A on the prediction is constrained. Let M denote that accepted mediator set; in the source-specific construction, M is chosen from the descendants of A :

$$M = \text{Desc}(A).$$

dCFF should not be described as equivalent to fairness through unawareness, because excluding A from the observed feature set does not generally eliminate proxy or mediated effects [21, 23].

Counterfactual equalised odds (CEqO). The decision D is independent of the protected attribute A conditional on the potential outcome $Y(1)$ under selection or treatment [25]:

$$D \perp A \mid Y(1).$$

The Pareto-dominance findings reported in [25] are theoretical limitations established under the assumptions of that analysis; they are not part of the definition itself.

Principal fairness (PF). The decision D is independent of A within principal strata defined jointly by the potential outcomes $Y(0)$ and $Y(1)$ [25]:

$$D \perp A \mid Y(0), Y(1).$$

Conditional principal fairness additionally conditions on a set of observed covariates W :

$$D \perp A \mid Y(0), Y(1), W.$$

B.2 Explainability/XAI metrics and evaluation protocols

Unlike fairness criteria, XAI evaluation combines mathematically defined metrics, statistical tests, benchmark procedures, and human-centred evaluation instruments. Accordingly, this section presents formal equations where a metric has an established mathematical definition and procedural specifications where evaluation depends on a testing, benchmarking, or user-evaluation protocol.

TCAV and concept sensitivity [26]

A Concept Activation Vector $v_c^{(l)} \in R^m$ is a normal vector to a linear separator trained in the activation space of layer l to distinguish examples of a user-defined concept C from counterexamples. For class k , define conceptual sensitivity as:

$$s(x; C, k, l) = \nabla h_k^{(l)} \left(f^{(l)}(x) \right)^\top v_c^{(l)}.$$

The corresponding TCAV score is:

$$TCAV(C, k, l) = \frac{|\{x \in X_k : s(x; C, k, l) > 0\}|}{|X_k|}.$$

The TCAV protocol repeats CAV construction with multiple random counterexample sets and tests the resulting TCAV scores against the null value 0.5. The number of repeated runs, the significance test, and any multiple-testing correction must be stated explicitly. The protocol does not establish a single universal number of runs or a universal acceptance threshold.

Infidelity and Max-Sensitivity [27]

For an explanation $\Phi(f, x)$, model f , input x , and perturbation I drawn from a specified distribution μ , explanation infidelity is:

$$Infid(\Phi, f, x; \mu) = E[(I^\top \Phi(f, x) - [f(x) - f(x - I)])^2], \quad I \sim \mu.$$

Lower values indicate lower expected mismatch for the specified perturbation distribution. The distribution μ must be reported because the interpretation of infidelity depends on it.

Define the local neighbourhood:

$$B(x, r) = \{y : \|y - x\| \leq r\}.$$

Max-Sensitivity is the largest change in an explanation within this neighbourhood:

$$MaxSens(\Phi, f, x; r) = \max\{\|\Phi(f, y) - \Phi(f, x)\| : y \in B(x, r)\}.$$

Lower values indicate greater local robustness, but Max-Sensitivity must be interpreted jointly with faithfulness because a trivial constant explanation can obtain low sensitivity. If a Lipschitz bound is reported, the corresponding local Lipschitz constant must be defined explicitly.

Quantus evaluation framework [28]

Quantus organises XAI evaluation into six categories. The interpretation direction is metric-specific:

1. **Faithfulness:** Infidelity ↓; Faithfulness Correlation ↑; and stronger predictive-performance degradation under a valid feature-removal protocol.
2. **Robustness:** generally ↓ for sensitivity or local-Lipschitz measures.
3. **Localisation:** generally ↑ where the metric evaluates overlap with a valid localisation target.
4. **Complexity:** generally ↓ for complexity measures; sparsity-type measures must be interpreted according to their exact definition.
5. **Randomisation:** larger explanation change, or lower similarity, after model or data randomisation is generally preferred.
6. **Axiomatic:** criterion-specific PASS/FAIL or conformity with the relevant axiom.

Category-level arrows must not replace metric-level interpretation. In particular, Infidelity is lower-is-better even though other faithfulness metrics may be higher-is-better.

Human-centred explanation evaluation [29]

The Explanation Satisfaction Scale evaluates perceived understanding, usefulness, sufficiency of detail, and related user-side qualities. Its psychometric reliability is reported through measures such as Cronbach's α , but the instrument does not define a universal pass/fail cut-off. EXCALIBUR should therefore compare explanation variants and stakeholder groups rather than treating a single satisfaction score as evidence of technical faithfulness.

The Explanation Goodness Checklist is an expert-facing quality check that can be applied before user exposure. Think-aloud protocols, mental-model measures, trust calibration, curiosity, and task-performance measures provide complementary evidence about whether an explanation supports appropriate user understanding.

Model and data randomisation tests [30]

For an explanation method $\mathcal{E}$, let f_θ be the trained model and f_θ a randomly reinitialised or progressively randomised version. Define

$$\mathcal{E}_T(x) = \mathcal{E}(x; f_\theta), \quad \mathcal{E}_R(x) = \mathcal{E}(x; f_\theta),$$

where T and R denote the trained and randomised cases. For a Data Randomisation Test, let D_p denote the training data with permuted labels and let $\mathcal{T}$ denote the training procedure. The retrained model is

$$f_p = \mathcal{T}(D_p).$$

Randomisation tests should use a pre-specified similarity measure and an explicitly documented acceptance rule. A method provides stronger evidence of dependence on learned parameters or labels when similarity decreases substantially after randomisation. The literature does not establish a universal numerical cut-off for “low similarity”; method-specific findings in [30] should not be presented as universal pass/fail properties across all models and datasets.

ROAR faithfulness benchmark [31]

Remove And Retrain (ROAR) evaluates whether features ranked as important carry predictive information:

- (1) **Rank features** by estimated importance using a protocol fixed before evaluation.
- (2) **Remove or mask features** in the top- t fraction using a domain-appropriate intervention.
- (3) **Retrain the model** from random initialisation on the modified training data.
- (4) **Evaluate predictive performance** on correspondingly modified test data.
- (5) **Compare the degradation curve** with random-ranking and other appropriate baselines.

The original benchmark was formulated for image features and used pixel replacement. Any application to tabular, wearable, or time-series data requires a separately justified feature-removal and retraining protocol.

B.3 Accountability instruments and process controls

SMACTR internal-audit framework [32]. The framework organises internal algorithmic auditing into five main phases—Scoping, Mapping, Artefact Collection, Testing, and Reflection—followed by post-audit follow-up. The phases produce traceable artefacts such as a project or product requirements document, stakeholder mapping, failure-mode analysis, model cards, dataset documentation, an adversarial-testing report, an Algorithmic Design History File, and an Audit Summary Report. The framework supports an internal go/no-go or remediation decision but does not replace an independent external audit.

For EXCALIBUR, a qualitative likelihood-by-severity table may be used as a project-defined prioritisation aid for identified failure modes. It must be labelled as an EXCALIBUR operational convention and must not be presented as a universally standardised FMEA equation or as a source-defined SMACTR threshold.

Datasheets for Datasets [16]. A dataset-level documentation instrument covering motivation, composition, collection, preprocessing or cleaning, recommended uses, distribution, and maintenance. Its value lies in structured creator reflection and traceability; no universal completeness score or certification threshold is defined.

Model Cards [15]. A model-level documentation instrument covering model details, intended use, relevant factors, evaluation metrics and data, training data, disaggregated quantitative analyses, ethical considerations, and caveats or recommendations. Presence alone does not establish documentation quality or legal compliance.

Responsible AI Metrics Catalogue [33]. The catalogue organises accountability metrics across responsibility, auditability, and redressability and distinguishes process, resource, and product metrics. In the reported

instantiation, process metrics are assessed through binary completion checks; the source identifies more graded scoring as future work. Any percentage-based “readiness” aggregation used by EXCALIBUR must therefore be labelled as a project-defined summary and must not be attributed to the source as a validated universal score.

For the purposes of EXCALIBUR, each applicable accountability process check may be represented by a project-defined binary completion variable:

$$c_j \in \{0,1\}, \quad j = 1, \dots, J$$

Here, $(c_j=1)$ indicates that the applicable process requirement (j) has been completed, evidenced, and reviewed, whereas $(c_j=0)$ indicates that the required evidence is absent or incomplete. The checks should be reported individually. No aggregate accountability or readiness score is inferred unless EXCALIBUR separately defines, documents, and justifies a project-specific aggregation rule.

Three-layered LLM audit blueprint [34]. Governance audits assess provider-level processes and institutional controls; model audits assess the released model’s performance, robustness, information security, truthfulness, and training-data evidence; and application audits assess functionality and real-world impacts in the specific deployment context. The three layers are complementary and should inform one another. For LLM-based EXCALIBUR components, a one-off model snapshot is insufficient; application-layer evaluation should combine ex-ante assessment with continuous ex-post monitoring.

B.4 Additional metrics and instruments surveyed but not carried forward

The following items were examined in the original literature corpus but were not selected for Tables 6–8. They remain relevant for later extensions, use-case-specific analysis, or methodological comparison.

Individual and causal fairness [21]

Conditional demographic disparity. Demographic parity evaluated within strata of legitimate explanatory variables.

Fairness through unawareness. The protected attribute is excluded from the predictor. This is an insufficient guarantee because other features may operate as proxies.

Fairness through awareness. Similar individuals should receive similar outcomes under an explicitly defined task-relevant similarity metric.

No unresolved discrimination. Every causal path from the protected attribute to the decision must pass through a resolving variable accepted as legitimate.

No proxy discrimination. The protected attribute has no causal effect on the decision through a variable functioning as a proxy.

Fairness index. A ratio-based summary comparing within-group and between-group fairness quantities; its interpretation depends on the underlying definition.

Additional causal and class-parity criteria [23, 25]

Interventional fairness. Fairness is evaluated under interventions on the protected attribute, conditional on a selected set of admissible variables.

Equal causal fairness. The causal effect of the protected attribute on the decision is constrained to zero under the specified model.

Counterfactual predictive parity. Predictive parity evaluated with respect to potential outcomes rather than only observed outcomes.

Conditional principal fairness. Principal fairness additionally conditioned on observed covariates.

Inframarginality. A diagnostic concept showing that group-level outcome or error-rate disparities can be misleading when risk distributions differ within groups.

Regression fairness [22]

Statistical parity for regression. The distribution of continuous predictions is constrained to be similar across protected groups.

Bounded group loss. The predictive loss of every protected group must remain below a specified bound.

Additional group and inequality diagnostics [20, 24]

Equalising disincentives. The difference between true-positive and false-positive rates is equalised across groups.

Conditional equal opportunity. Equal opportunity is evaluated within strata of an auxiliary legitimate attribute.

Bayesian fairness. Fairness is evaluated over the posterior distribution of model parameters rather than at a single point estimate.

Generalised entropy index. A population inequality measure over individual benefits that can be decomposed into between-group and within-group components.

Theil index. The $\alpha = 1$ case of the generalised entropy family; it is additively decomposable into between-group and within-group inequality.

Additional explainability instruments [27, 29]

Smoothed explanations (SmoothGrad family). Kernel smoothing may reduce both infidelity and local sensitivity under the assumptions analysed in [27].

Explanation Goodness Checklist. An expert-facing, a priori quality checklist applied before explanations are presented to users.

Mental-model, trust, and curiosity instruments. Think-aloud protocols, curiosity measures, trust calibration, and multi-level performance measures for human-centred explanation evaluation.

B.5 Supplementary sources added in the updated deliverable

The following sources were introduced during the final update of D2.1 and were not part of the original 18-paper P001–P018 corpus. They are listed separately to preserve the provenance of the original literature review while maintaining complete traceability for the updated narrative and tables.

Uncertainty-aware fairness for LLMs [41]. UCerF incorporates predictive uncertainty into fairness evaluation so that equally sized error-rate gaps can be interpreted differently when model certainty differs. It is an emerging research metric and does not provide a universal numerical threshold.

Healthcare LLM bias-evaluation framework [42]. The framework structures bias evaluation around stakeholder and use-case definition, model and data characterisation, scenario-based testing, calibration or uncertainty assessment, and interpretation of output-level harms. It is a domain-specific audit framework rather than a general compliance checklist.

Survey of LLM evaluation [43]. The survey organises evaluation across capabilities, robustness, bias, safety, and related dimensions and emphasises the use of both automatic benchmarks and human evaluation. It supports the multidimensional evaluation logic adopted in Sections 6 and 7.

Influence functions and training-data attribution [44]. Influence functions estimate how individual training examples affect a model prediction or parameter estimate. They provide data-level provenance evidence but depend on modelling assumptions and may be computationally difficult to apply at large-model scale.

Together with Appendix A, Sections B.1–B.4 provide traceability for the metrics, evaluation frameworks, and accountability instruments extracted from the original 18-paper Task 2.1 corpus (P001–P018). Section B.5 records supplementary sources [41]–[44] introduced during the final update of the deliverable. This separation preserves the provenance of the original review while maintaining a complete audit trail from the updated main text to the corresponding technical or governance source.

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